

Exploratory Data Analysis

```
In [42]: import os
import warnings

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns

from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.seasonal import STL
from statsmodels.tsa.stattools import adfuller, kpss

ROOT_PATH = os.path.dirname(os.getcwd())
DATA_FOLDER_PATH = os.path.join(ROOT_PATH, 'data', 'processed')

COVID_START = pd.Timestamp('2020-04-01') # first pandemic month: April 2020
COVID_END = pd.Timestamp('2021-12-31')

sns.set_palette('colorblind')
```

Helper Functions — Modern Time Series Forecasting with Python (2E)

The trend, seasonality, heteroscedasticity and transformation checks below use helper functions from the companion repository of:

Joseph, M. & Tackes, J. *Modern Time Series Forecasting with Python*, 2nd ed.
(Packt, 2024) — [PacktPublishing/Modern-Time-Series-Forecasting-with-Python-2E](#), MIT License

Helper	Source file	Used for
check_trend	src/transforms/stationary_utils.py	Kendall's τ Mann-Kendall trend tests
check_deterministic_trend	src/transforms/stationary_utils.py	ADF (c vs ct) deterministic trend check
check_seasonality	src/transforms/stationary_utils.py	ACF + Bartlett significance test (adapted by the author from <code>dart</code>)

Helper	Source file	Used for
<code>check_heteroscedasticity</code>	<code>src/transforms/stationary_utils.py</code>	White test linear-trend
<code>DetrendingTransformer</code>	<code>src/transforms/target_transformations.py</code>	Linear detrending
<code>DeseasonalizingTransformer</code>	<code>src/transforms/target_transformations.py</code>	Period-average deseasonal
<code>BoxCoxTransformer</code>	<code>src/transforms/target_transformations.py</code>	Box-Cox w Guerrero λ (adapted b the author: from <code>sktime</code>
<code>AutoStationaryTransformer</code>	<code>src/transforms/target_transformations.py</code>	Automatic detrend → deseasonal → Box-Cox pipeline

The repository is expected as a sibling folder of this project (override with the `MTSF_REPO_PATH` environment variable). Its `src` package needs `plotly` and `pymannkendall`, both listed in `pyproject.toml`.

```
In [43]: import sys

# Helpers credited to: Joseph, M. & Tackes, J. "Modern Time Series Forecasting with
# Python", 2nd ed. (Packt, 2024) – MIT License
# github.com/PacktPublishing/Modern-Time-Series-Forecasting-with-Python-2E
MTSF_REPO_PATH = os.getenv(
    'MTSF_REPO_PATH',
    os.path.join(
        os.path.dirname(ROOT_PATH), 'Modern-Time-Series-Forecasting-with-Python-2E'
    ),
)
if MTSF_REPO_PATH not in sys.path:
    sys.path.insert(0, MTSF_REPO_PATH)

from src.transforms.stationary_utils import ( # noqa: E402
    check_deterministic_trend,
    check_heteroscedasticity,
    check_seasonality,
    check_trend,
)
from src.transforms.target_transformations import ( # noqa: E402
    AutoStationaryTransformer,
    BoxCoxTransformer,
    DeseasonalizingTransformer,
    DetrendingTransformer,
)
```

```
SEASONAL_PERIOD = 12
FREQ = 'ME'
```

Data Import

The EDA uses **train + validation** (`train.parquet` and `val.parquet` , notebook 2). The test set (the last 12 months) is never read here: it is kept for the final evaluation only.

```
In [44]: df = pd.concat(
    [
        pd.read_parquet(os.path.join(DATA_FOLDER_PATH, file_name))
        for file_name in ['train.parquet', 'val.parquet']
    ],
    ignore_index=True,
)

df = df.set_index('date')
df['is_covid_19'] = (df.index >= COVID_START) & (df.index <= COVID_END)

df.head()
```

```
Out[44]:
```

	access_route	state	origin_country	month	year	arrival_count	month_num	y
	date							
1989-01-31	Aérea	Amazonas	África do Sul	Janeiro	1989	9.0	01	
1989-01-31	Aérea	Amazonas	Angola	Janeiro	1989	0.0	01	
1989-01-31	Aérea	Amazonas	Nigéria	Janeiro	1989	0.0	01	
1989-01-31	Aérea	Amazonas	Outros países	Janeiro	1989	0.0	01	
1989-01-31	Aérea	Amazonas	Costa Rica	Janeiro	1989	6.0	01	

Categorical Features Analysis

Access Route

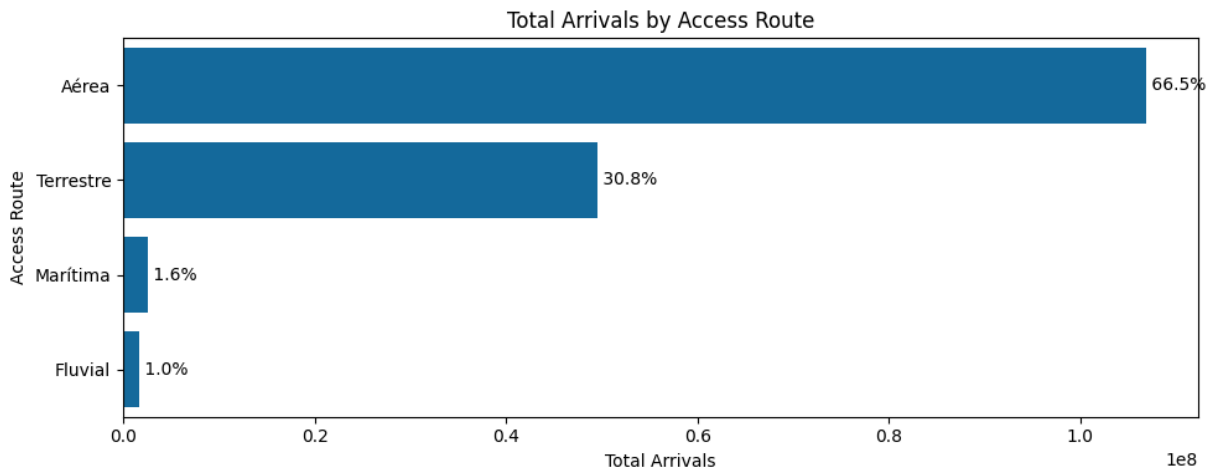
```
In [45]: top_access_route = pd.DataFrame(
    df.groupby('access_route')['arrival_count'].sum().nlargest(20)
)
top_access_route['percentage'] = (
    top_access_route['arrival_count'] / top_access_route['arrival_count'].sum()
) * 100
```

```
plt.figure(figsize=(10, 4))

sns.barplot(x=top_access_route['arrival_count'], y=top_access_route.index)
plt.title('Total Arrivals by Access Route')
plt.xlabel('Total Arrivals')
plt.ylabel('Access Route')

ax = plt.gca()
for i, (count, pct) in enumerate(
    zip(top_access_route['arrival_count'], top_access_route['percentage'])):
    ax.text(count, i, f' {pct:.1f}%', va='center', ha='left')

plt.tight_layout()
```



```
In [46]: route_time = (
    df[df.index.year >= 2016]
    .groupby(['year_month', 'access_route'], as_index=False)['arrival_count']
    .sum()
)

fig, ax = plt.subplots(figsize=(14, 5))

sns.lineplot(
    data=route_time,
    x='year_month',
    y='arrival_count',
    hue='access_route',
    linewidth=2,
    ax=ax,
)

ax.axvline(COVID_START, color='gray', linestyle=':', linewidth=1.5, label='_nolegen')
ax.axvline(COVID_END, color='gray', linestyle=':', linewidth=1.5, label='_nolegend')
ax.axvspan(COVID_START, COVID_END, alpha=0.07, color='gray')

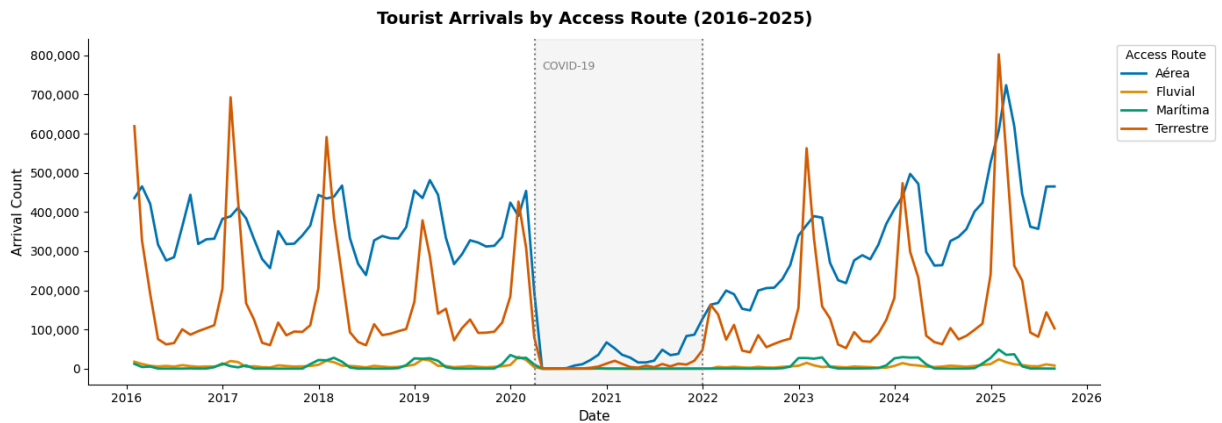
y_max = route_time['arrival_count'].max()
ax.text(
    COVID_START + pd.Timedelta(days=30),
    y_max * 0.95,
    'COVID-19',
```

```

        fontsize=9,
        color='gray',
    )

    ax.set_title(
        f'Tourist Arrivals by Access Route (2016-{df.index.year.max()})',
        fontsize=14,
        fontweight='bold',
        pad=12,
    )
    ax.set_xlabel('Date', fontsize=11)
    ax.set_ylabel('Arrival Count', fontsize=11)
    ax.legend(
        title='Access Route', bbox_to_anchor=(1.01, 1), loc='upper left', frameon=True
    )
    ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
    sns.despine()
    plt.tight_layout()

```



```

In [47]: month_route = (
    df.groupby(['month_num', 'access_route'])['arrival_count'].mean().reset_index()
)

month_totals = month_route.groupby('month_num')['arrival_count'].sum()
month_route['pct'] = month_route.apply(
    lambda r: r['arrival_count'] / month_totals[r['month_num']] * 100, axis=1
)

month_labels = [
    'Jan',
    'Feb',
    'Mar',
    'Apr',
    'May',
    'Jun',
    'Jul',
    'Aug',
    'Sep',
    'Oct',
    'Nov',
    'Dec',
]

```

```

month_route['month_label'] = (
    month_route['month_num'].astype(int).apply(lambda m: month_labels[m - 1])
)

fig, ax = plt.subplots(figsize=(16, 6))

sns.barplot(
    data=month_route,
    x='month_label',
    y='arrival_count',
    hue='access_route',
    order=month_labels,
    ax=ax,
)

for bar in ax.patches:
    height = bar.get_height()
    if height > 0:
        x = bar.get_x() + bar.get_width() / 2
        ax.text(
            x,
            height + 0.5,
            f'{height / month_route["arrival_count"].max() * 100:.0f}%',
            ha='center',
            va='bottom',
            fontsize=6.5,
            color='dimgray',
        )

# re-annotate with actual pct values per bar group
ax.cla()
bar_plot = sns.barplot(
    data=month_route,
    x='month_label',
    y='arrival_count',
    hue='access_route',
    order=month_labels,
    ax=ax,
)

routes = month_route['access_route'].unique()
for month_idx, month in enumerate(month_labels):
    month_data = month_route[month_route['month_label'] == month].set_index(
        'access_route'
    )
    n_routes = len(routes)
    total_width = 0.8
    bar_width = total_width / n_routes
    for route_idx, route in enumerate(routes):
        if route not in month_data.index:
            continue
        row = month_data.loc[route]
        x = month_idx - total_width / 2 + bar_width * route_idx + bar_width / 2
        ax.text(
            x,
            row['arrival_count'] + 0.3,

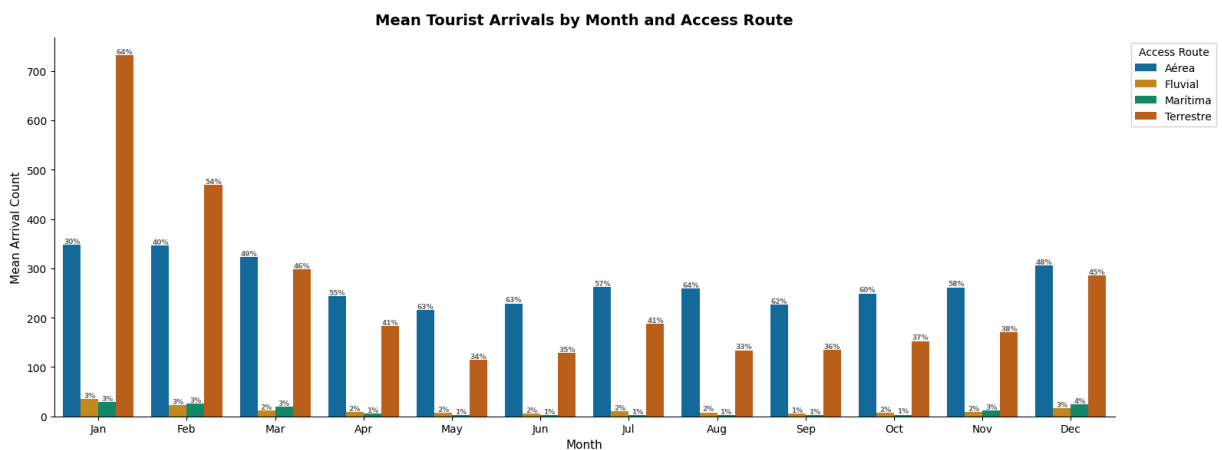
```

```

        f'{row["pct"]:.0f}%',
        ha='center',
        va='bottom',
        fontsize=6.5,
        color='dimgray',
        fontweight='bold',
    )

ax.set_title(
    'Mean Tourist Arrivals by Month and Access Route',
    fontsize=14,
    fontweight='bold',
    pad=12,
)
ax.set_xlabel('Month', fontsize=11)
ax.set_ylabel('Mean Arrival Count', fontsize=11)
ax.legend(
    title='Access Route', bbox_to_anchor=(1.01, 1), loc='upper left', frameon=True
)
ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
sns.despine()
plt.tight_layout()

```



Arrival State and Region

Arrival State

```

In [48]: top_states = pd.DataFrame(df.groupby('state')['arrival_count'].sum().nlargest(30))
top_states['percentage'] = (
    top_states['arrival_count'] / top_states['arrival_count'].sum()
) * 100

plt.figure(figsize=(10, 6))

sns.barplot(x=top_states['arrival_count'], y=top_states.index)
plt.title('Total Arrivals by Destination State')
plt.xlabel('Total Arrivals')
plt.ylabel('State')

ax = plt.gca()

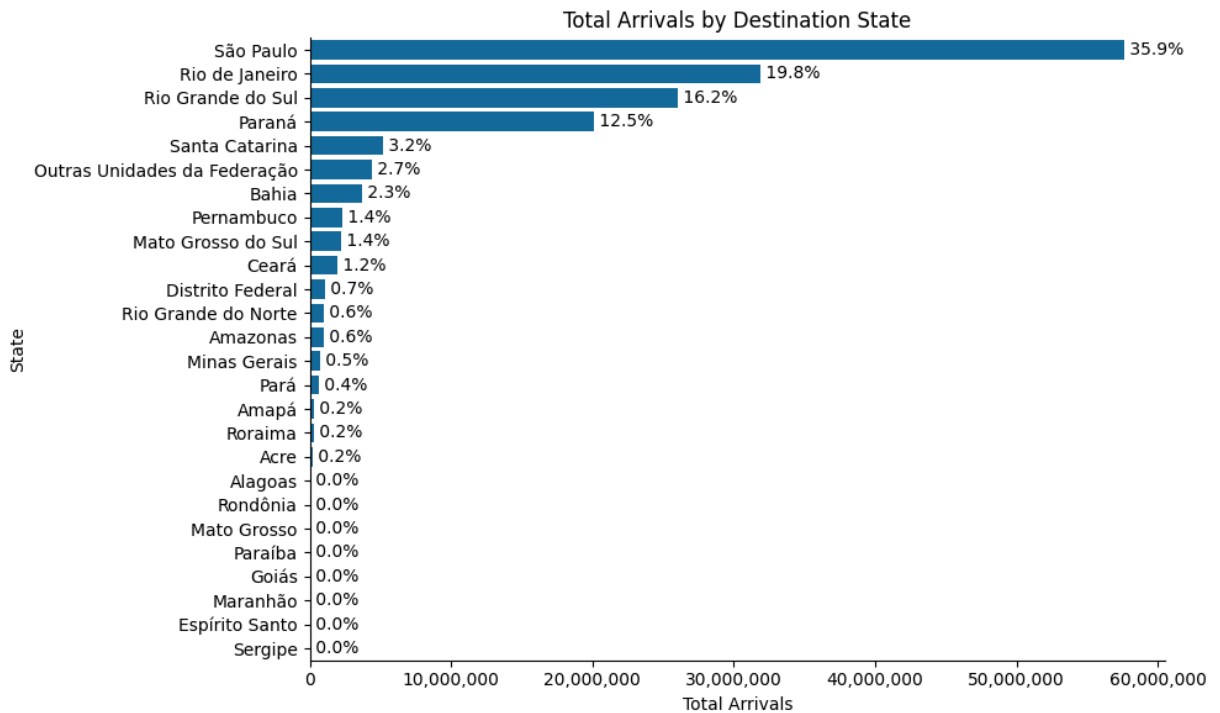
```

```

for i, (count, pct) in enumerate(
    zip(top_states['arrival_count'], top_states['percentage']))
):
    ax.text(count, i, f' {pct:.1f}%', va='center', ha='left')

ax.xaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
sns.despine()
plt.tight_layout()

```



```

In [49]: TOP_N = 8
top_state_names = (
    df.groupby('state')['arrival_count'].sum().nlargest(TOP_N).index.tolist()
)

state_time = (
    df[df['state'].isin(top_state_names) & (df.index.year >= 2016)]
    .groupby(['year_month', 'state'], as_index=False)['arrival_count']
    .sum()
)

fig, ax = plt.subplots(figsize=(14, 5))

sns.lineplot(
    data=state_time,
    x='year_month',
    y='arrival_count',
    hue='state',
    linewidth=2,
    ax=ax,
)

ax.axvline(COVID_START, color='gray', linestyle=':', linewidth=1.5, label='_nolegend')
ax.axvline(COVID_END, color='gray', linestyle=':', linewidth=1.5, label='_nolegend')
ax.axvspan(COVID_START, COVID_END, alpha=0.07, color='gray')

```

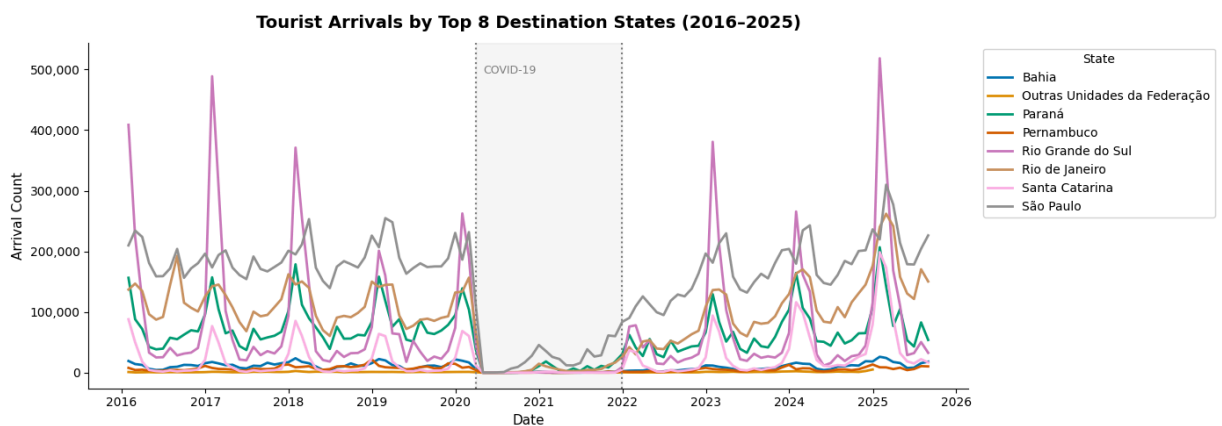
```

y_max = state_time['arrival_count'].max()
ax.text(
    COVID_START + pd.Timedelta(days=30),
    y_max * 0.95,
    'COVID-19',
    fontsize=9,
    color='gray',
)

ax.set_title(
    f'Tourist Arrivals by Top {TOP_N} Destination States (2016-{df.index.year.max()})',
    fontsize=14,
    fontweight='bold',
    pad=12,
)

ax.set_xlabel('Date', fontsize=11)
ax.set_ylabel('Arrival Count', fontsize=11)
ax.legend(title='State', bbox_to_anchor=(1.01, 1), loc='upper left', frameon=True)
ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
sns.despine()
plt.tight_layout()

```



```

In [50]: month_labels = [
    'Jan',
    'Feb',
    'Mar',
    'Apr',
    'May',
    'Jun',
    'Jul',
    'Aug',
    'Sep',
    'Oct',
    'Nov',
    'Dec',
]

state_month = (
    df[df['state'].isin(top_state_names)]
    .groupby(['state', 'month_num'])['arrival_count']
    .mean()
)

```

```

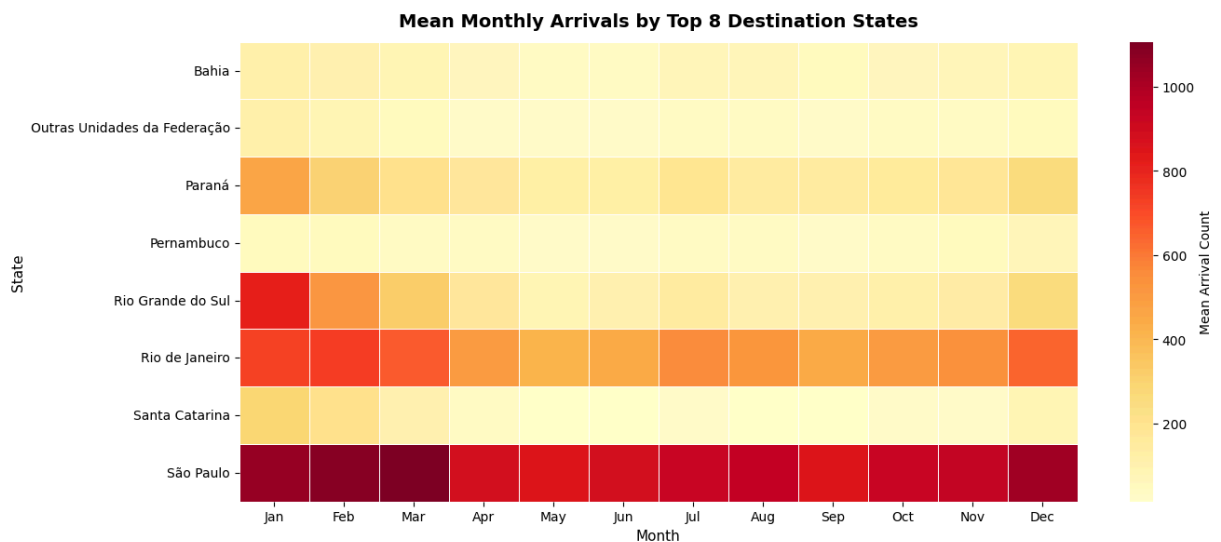
        .unstack('month_num')
    )
    state_month.columns = month_labels

    fig, ax = plt.subplots(figsize=(14, 6))

    sns.heatmap(
        state_month,
        fmt='.0f',
        linewidths=0.4,
        cmap='YlOrRd',
        ax=ax,
        cbar_kws={'label': 'Mean Arrival Count'},
    )

    ax.set_title(
        f'Mean Monthly Arrivals by Top {TOP_N} Destination States',
        fontsize=14,
        fontweight='bold',
        pad=12,
    )
    ax.set_xlabel('Month', fontsize=11)
    ax.set_ylabel('State', fontsize=11)
    ax.tick_params(axis='y', rotation=0)
    plt.tight_layout()

```



Arrival Region

```

In [51]: region_totals = pd.DataFrame(
    df.groupby('state_region')['arrival_count'].sum().sort_values(ascending=False)
)
region_totals['percentage'] = (
    region_totals['arrival_count'] / region_totals['arrival_count'].sum()
) * 100

plt.figure(figsize=(10, 4))

sns.barplot(x=region_totals['arrival_count'], y=region_totals.index)

```

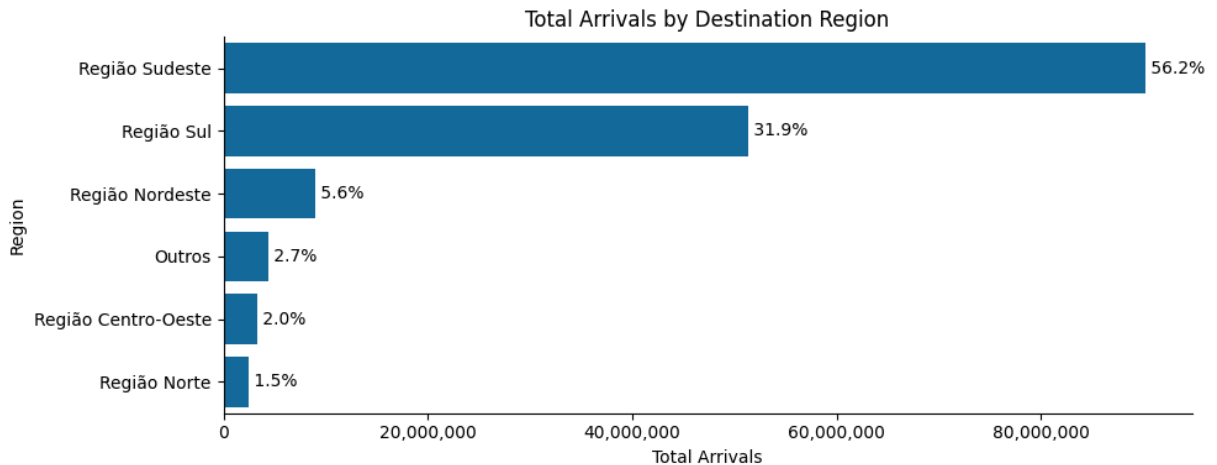
```

plt.title('Total Arrivals by Destination Region')
plt.xlabel('Total Arrivals')
plt.ylabel('Region')

ax = plt.gca()
for i, (count, pct) in enumerate(
    zip(region_totals['arrival_count'], region_totals['percentage'])):
    ax.text(count, i, f' {pct:.1f}%', va='center', ha='left')

ax.xaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
sns.despine()
plt.tight_layout()

```



```

In [52]: region_time = (
    df[df.index.year >= 2016]
    .groupby(['year_month', 'state_region'], as_index=False)['arrival_count']
    .sum()
)

fig, ax = plt.subplots(figsize=(14, 5))

sns.lineplot(
    data=region_time,
    x='year_month',
    y='arrival_count',
    hue='state_region',
    linewidth=2,
    ax=ax,
)

ax.axvline(COVID_START, color='gray', linestyle=':', linewidth=1.5, label='_nolegen
ax.axvline(COVID_END, color='gray', linestyle=':', linewidth=1.5, label='_nolegend_
ax.axvspan(COVID_START, COVID_END, alpha=0.07, color='gray')

y_max = region_time['arrival_count'].max()
ax.text(
    COVID_START + pd.Timedelta(days=30),
    y_max * 0.95,
    'COVID-19',
    fontsize=9,

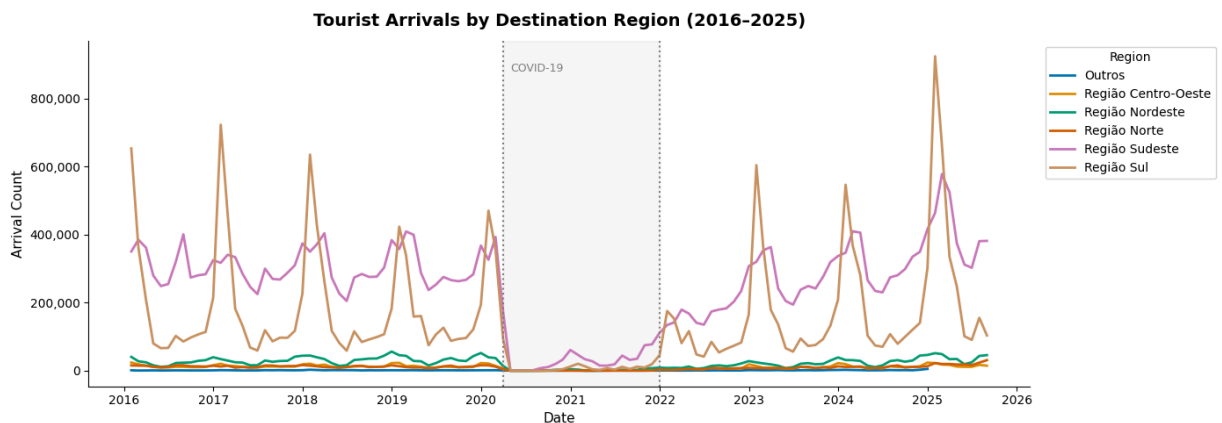
```

```

        color='gray',
    )

    ax.set_title(
        f'Tourist Arrivals by Destination Region (2016-{df.index.year.max()})',
        fontsize=14,
        fontweight='bold',
        pad=12,
    )
    ax.set_xlabel('Date', fontsize=11)
    ax.set_ylabel('Arrival Count', fontsize=11)
    ax.legend(title='Region', bbox_to_anchor=(1.01, 1), loc='upper left', frameon=True)
    ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
    sns.despine()
    plt.tight_layout()

```



```

In [53]: month_labels = [
    'Jan',
    'Feb',
    'Mar',
    'Apr',
    'May',
    'Jun',
    'Jul',
    'Aug',
    'Sep',
    'Oct',
    'Nov',
    'Dec',
]

region_month = (
    df
    .groupby(['state_region', 'month_num'])['arrival_count']
    .mean()
    .unstack('month_num')
)
region_month.columns = month_labels

fig, ax = plt.subplots(figsize=(14, 5))

sns.heatmap(

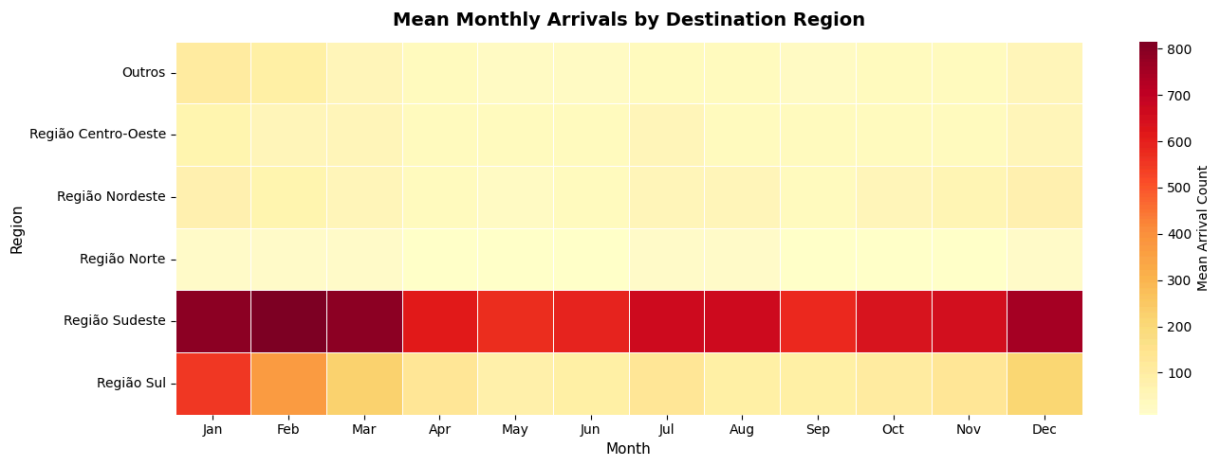
```

```

    region_month,
    fmt='.0f',
    linewidths=0.4,
    cmap='YlOrRd',
    ax=ax,
    cbar_kws={'label': 'Mean Arrival Count'},
)

ax.set_title(
    'Mean Monthly Arrivals by Destination Region',
    fontsize=14,
    fontweight='bold',
    pad=12,
)
ax.set_xlabel('Month', fontsize=11)
ax.set_ylabel('Region', fontsize=11)
ax.tick_params(axis='y', rotation=0)
plt.tight_layout()

```



Arrival Country / Cultural Subdivision

Origin Country

```

In [54]: TOP_N_COUNTRIES = 20

top_countries = pd.DataFrame(
    df.groupby('origin_country')['arrival_count'].sum().nlargest(TOP_N_COUNTRIES)
)
top_countries['percentage'] = (
    top_countries['arrival_count']
    / df.groupby('origin_country')['arrival_count'].sum().sum()
) * 100

plt.figure(figsize=(10, 7))

sns.barplot(x=top_countries['arrival_count'], y=top_countries.index)
plt.title(f'Total Arrivals by Top {TOP_N_COUNTRIES} Origin Countries')
plt.xlabel('Total Arrivals')
plt.ylabel('Origin Country')

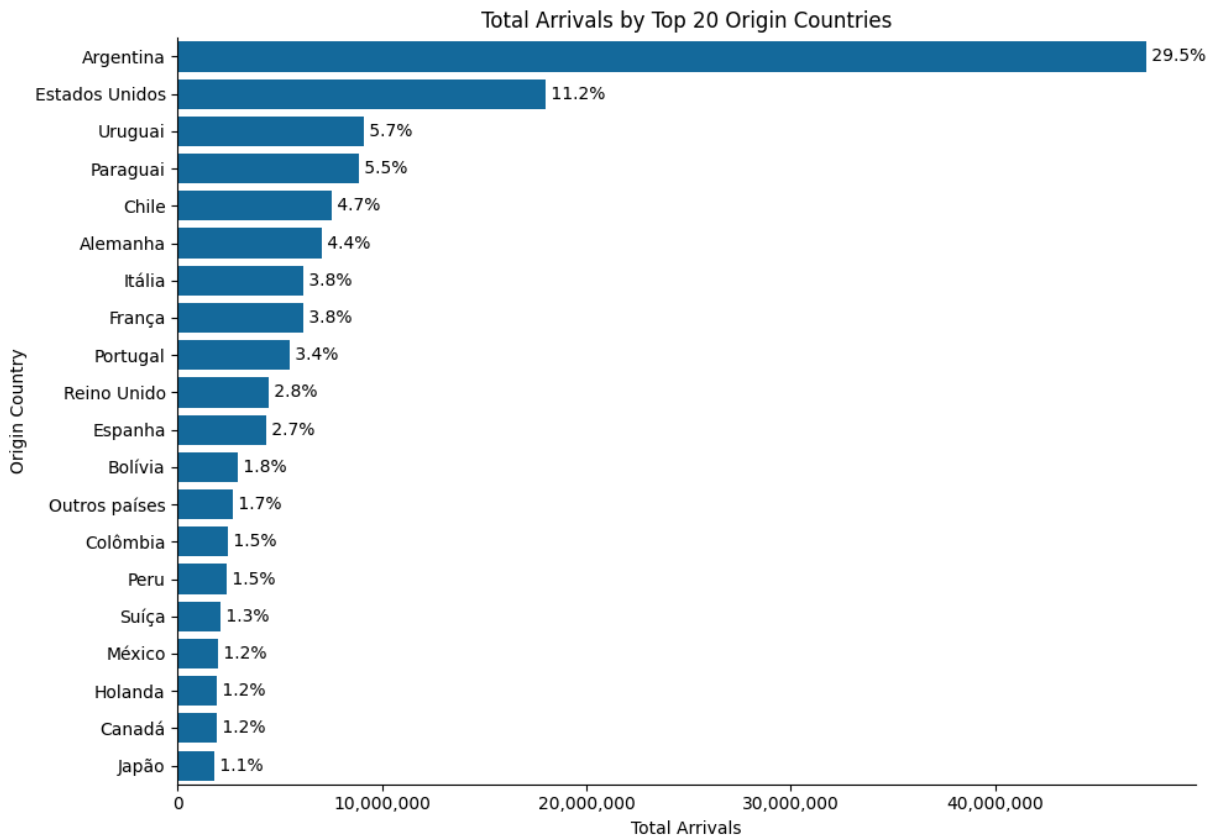
```

```

ax = plt.gca()
for i, (count, pct) in enumerate(
    zip(top_countries['arrival_count'], top_countries['percentage'])):
    ax.text(count, i, f' {pct:.1f}%', va='center', ha='left')

ax.xaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
sns.despine()
plt.tight_layout()

```



In [55]: TOP_N_COUNTRIES = 10

```

top_country_names = (
    df
    .groupby('origin_country')['arrival_count']
    .sum()
    .nlargest(TOP_N_COUNTRIES)
    .index.tolist()
)

country_time = (
    df[df['origin_country'].isin(top_country_names) & (df.index.year >= 2016)]
    .groupby(['year_month', 'origin_country'], as_index=False)['arrival_count']
    .sum()
)

fig, ax = plt.subplots(figsize=(14, 5))

sns.lineplot(
    data=country_time,

```

```

x='year_month',
y='arrival_count',
hue='origin_country',
linewidth=2,
ax=ax,
)

ax.axvline(COVID_START, color='gray', linestyle=':', linewidth=1.5, label='_nolegen')
ax.axvline(COVID_END, color='gray', linestyle=':', linewidth=1.5, label='_nolegend_')
ax.axvspan(COVID_START, COVID_END, alpha=0.07, color='gray')

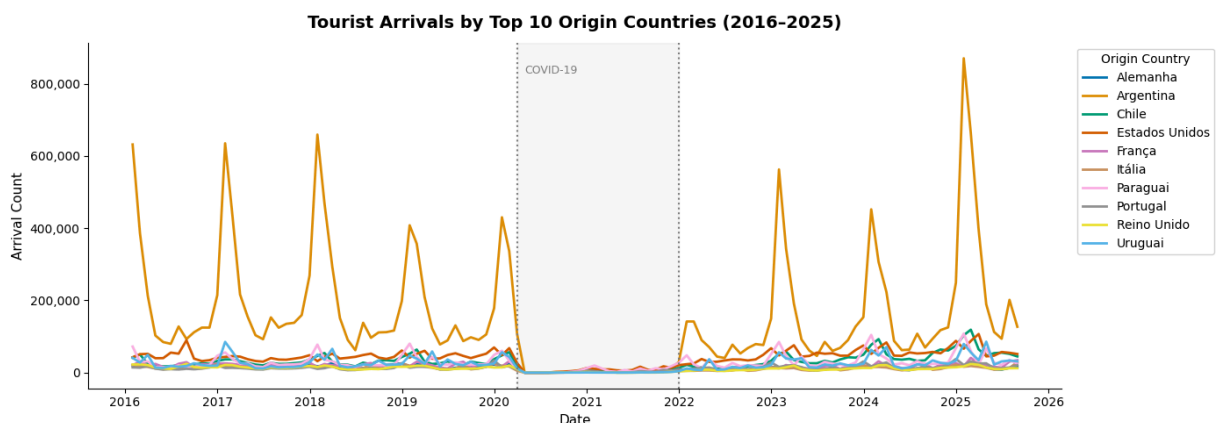
y_max = country_time['arrival_count'].max()
ax.text(
    COVID_START + pd.Timedelta(days=30),
    y_max * 0.95,
    'COVID-19',
    fontsize=9,
    color='gray',
)

ax.set_title(
    f'Tourist Arrivals by Top {TOP_N_COUNTRIES} Origin Countries '
    f'(2016-{df.index.year.max()})',
    fontsize=14,
    fontweight='bold',
    pad=12,
)

ax.set_xlabel('Date', fontsize=11)
ax.set_ylabel('Arrival Count', fontsize=11)
ax.legend(
    title='Origin Country', bbox_to_anchor=(1.01, 1), loc='upper left', frameon=True
)

ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
sns.despine()
plt.tight_layout()

```



```

In [56]: month_labels = [
    'Jan',
    'Feb',
    'Mar',
    'Apr',
    'May',

```

```

    'Jun',
    'Jul',
    'Aug',
    'Sep',
    'Oct',
    'Nov',
    'Dec',
]

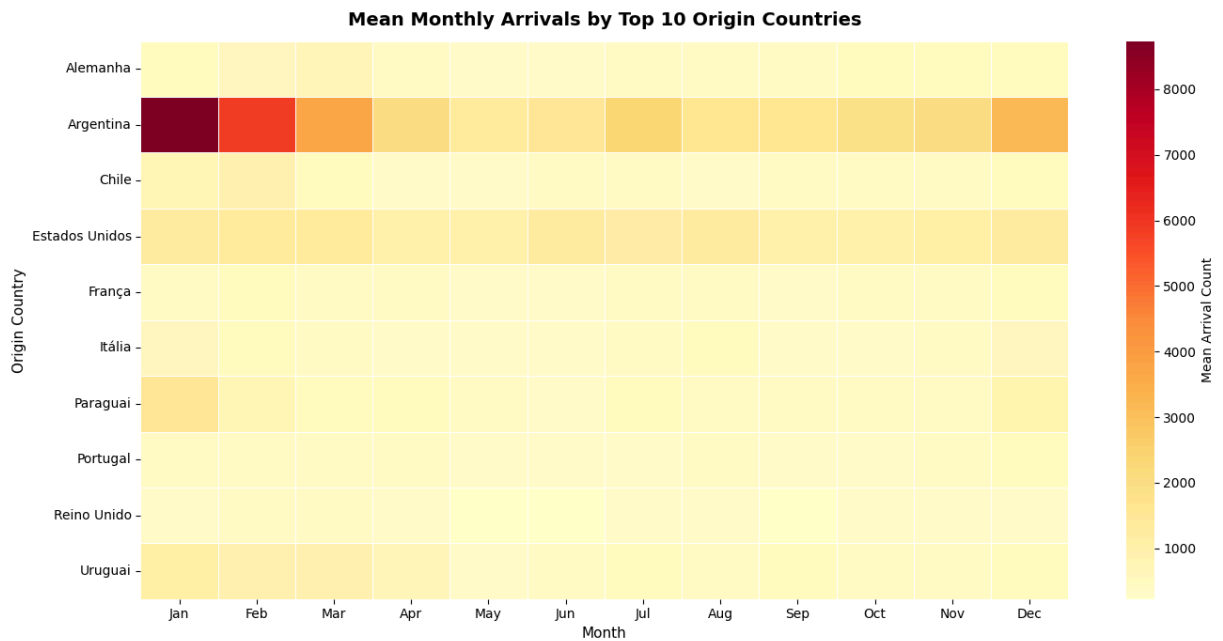
country_month = (
    df[df['origin_country'].isin(top_country_names)]
    .groupby(['origin_country', 'month_num'])['arrival_count']
    .mean()
    .unstack('month_num')
)
country_month.columns = month_labels

fig, ax = plt.subplots(figsize=(14, 7))

sns.heatmap(
    country_month,
    fmt='.0f',
    linewidths=0.4,
    cmap='YlOrRd',
    ax=ax,
    cbar_kws={'label': 'Mean Arrival Count'},
)

ax.set_title(
    f'Mean Monthly Arrivals by Top {TOP_N_COUNTRIES} Origin Countries',
    fontsize=14,
    fontweight='bold',
    pad=12,
)
ax.set_xlabel('Month', fontsize=11)
ax.set_ylabel('Origin Country', fontsize=11)
ax.tick_params(axis='y', rotation=0)
plt.tight_layout()

```



Cultural Subdivision

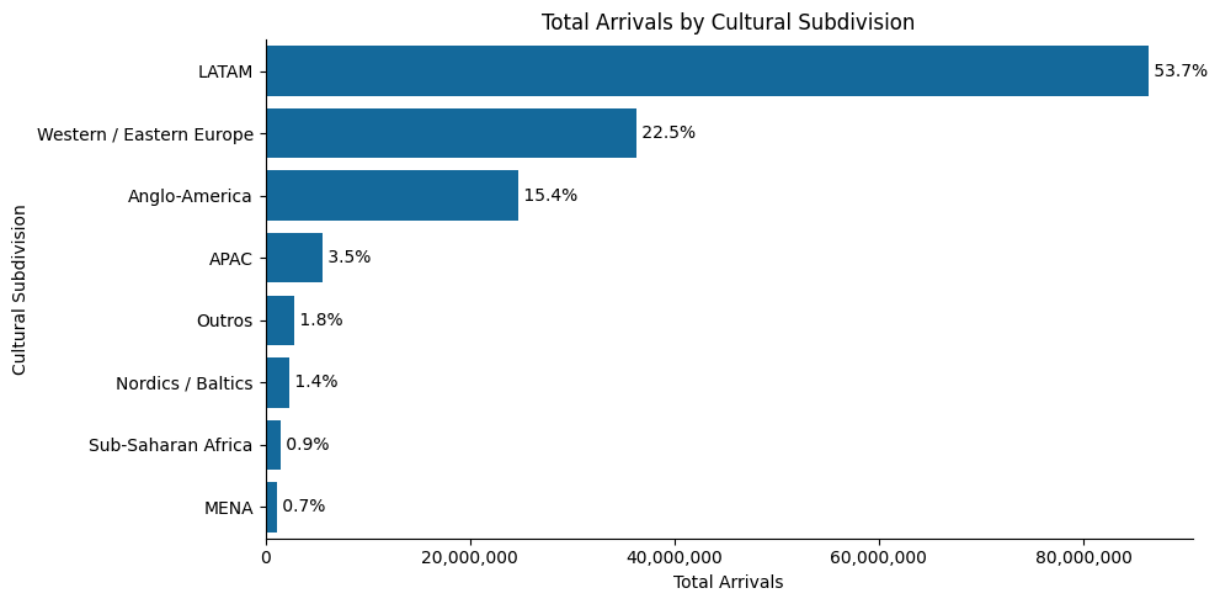
```
In [57]: subdivision_totals = pd.DataFrame(
    df
    .groupby('country_cultural_subdivision')['arrival_count']
    .sum()
    .sort_values(ascending=False)
)
subdivision_totals['percentage'] = (
    subdivision_totals['arrival_count'] / subdivision_totals['arrival_count'].sum()
) * 100

plt.figure(figsize=(10, 5))

sns.barplot(x=subdivision_totals['arrival_count'], y=subdivision_totals.index)
plt.title('Total Arrivals by Cultural Subdivision')
plt.xlabel('Total Arrivals')
plt.ylabel('Cultural Subdivision')

ax = plt.gca()
for i, (count, pct) in enumerate(
    zip(subdivision_totals['arrival_count'], subdivision_totals['percentage'])
):
    ax.text(count, i, f' {pct:.1f}%', va='center', ha='left')

ax.xaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
sns.despine()
plt.tight_layout()
```



```
In [58]: subdivision_time = (
    df[df.index.year >= 2016]
    .groupby(['year_month', 'country_cultural_subdivision'], as_index=False)[
        'arrival_count'
    ]
    .sum()
)

fig, ax = plt.subplots(figsize=(14, 5))

sns.lineplot(
    data=subdivision_time,
    x='year_month',
    y='arrival_count',
    hue='country_cultural_subdivision',
    linewidth=2,
    ax=ax,
)

ax.axvline(COVID_START, color='gray', linestyle=':', linewidth=1.5, label='_nolegen')
ax.axvline(COVID_END, color='gray', linestyle=':', linewidth=1.5, label='_nolegend_')
ax.axvspan(COVID_START, COVID_END, alpha=0.07, color='gray')

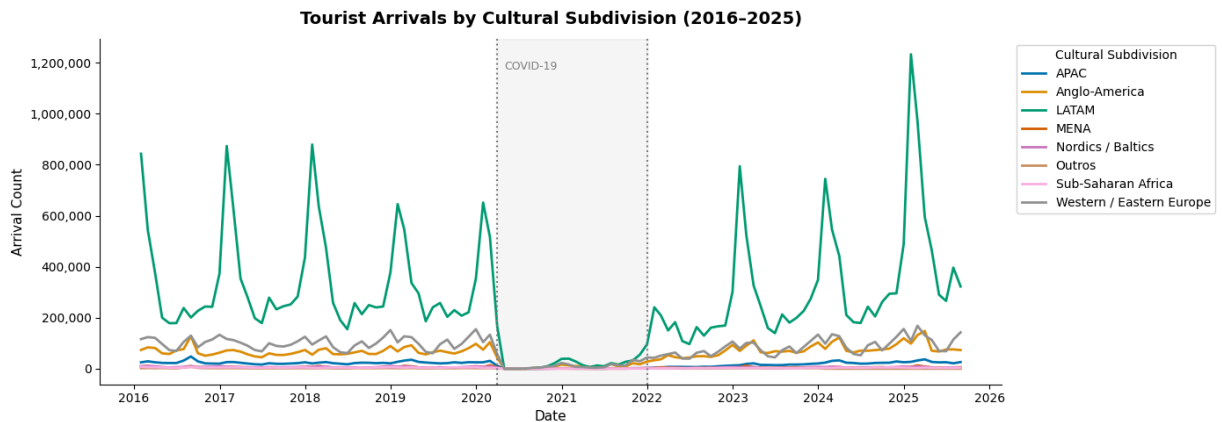
y_max = subdivision_time['arrival_count'].max()
ax.text(
    COVID_START + pd.Timedelta(days=30),
    y_max * 0.95,
    'COVID-19',
    fontsize=9,
    color='gray',
)

ax.set_title(
    f'Tourist Arrivals by Cultural Subdivision (2016-{df.index.year.max()})',
    fontsize=14,
    fontweight='bold',
    pad=12,
)
```

```

)
ax.set_xlabel('Date', fontsize=11)
ax.set_ylabel('Arrival Count', fontsize=11)
ax.legend(
    title='Cultural Subdivision',
    bbox_to_anchor=(1.01, 1),
    loc='upper left',
    frameon=True,
)
ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
sns.despine()
plt.tight_layout()

```



```

In [59]: month_labels = [
    'Jan',
    'Feb',
    'Mar',
    'Apr',
    'May',
    'Jun',
    'Jul',
    'Aug',
    'Sep',
    'Oct',
    'Nov',
    'Dec',
]

subdivision_month = (
    df
    .groupby(['country_cultural_subdivision', 'month_num'])['arrival_count']
    .mean()
    .unstack('month_num')
)
subdivision_month.columns = month_labels

fig, ax = plt.subplots(figsize=(14, 5))

sns.heatmap(
    subdivision_month,
    fmt='%.0f',
    linewidths=0.4,

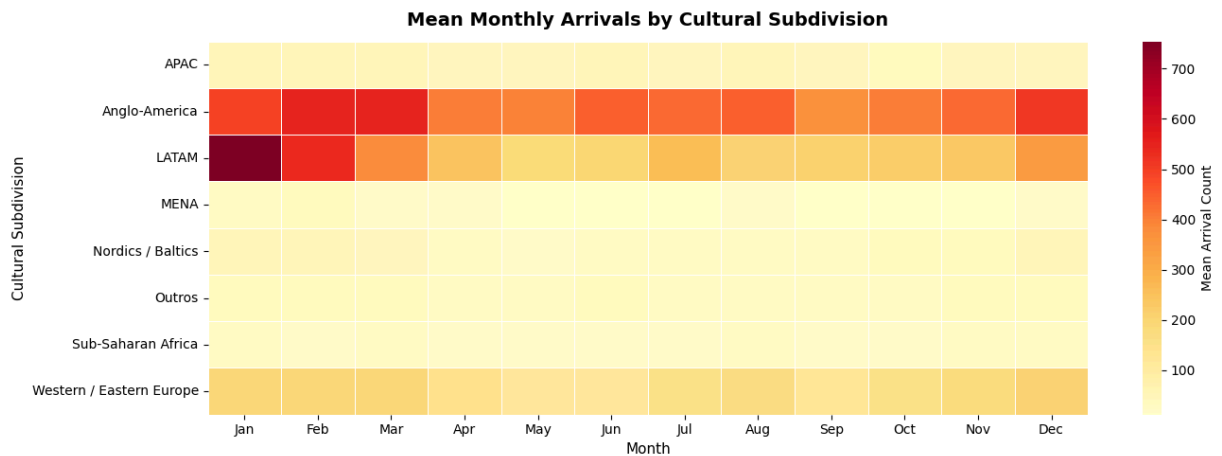
```

```

cmap='YlOrRd',
ax=ax,
cbar_kws={'label': 'Mean Arrival Count'},
)

ax.set_title(
    'Mean Monthly Arrivals by Cultural Subdivision',
    fontsize=14,
    fontweight='bold',
    pad=12,
)
ax.set_xlabel('Month', fontsize=11)
ax.set_ylabel('Cultural Subdivision', fontsize=11)
ax.tick_params(axis='y', rotation=0)
plt.tight_layout()

```



Time Series Analysis

Note on transformations. An earlier version of this section applied `np.log` to the series before the additive/multiplicative check, the STL decomposition, the ADF/KPSS tests and the ACF/PACF plots. The log was there for two reasons:

1. **Multiplicative** → **additive**. The mean–std check shows the seasonal amplitude growing with the level (multiplicative behaviour). STL is additive, so the log was used to turn `trend × season × error` into `trend + season + error`.
2. **Variance stabilization**. A log damps the level-dependent variance before running unit-root tests and reading autocorrelations.

The log is the fixed special case $\lambda = 0$ of the Box-Cox family, and it was applied without first testing whether the variance actually changes with the level, or whether $\lambda = 0$ is the right strength. The sections below therefore work on the **raw scale**. Heteroscedasticity is then tested formally and λ is chosen by **Guerrero's method** (see *Heteroscedasticity* and *Box-Cox Transformation*).

Results on the raw scale should be read with that in mind: the STL seasonal component absorbs the growing amplitude, and ADF/KPSS results can be influenced by the changing variance.

Overview

```
In [60]: monthly_df = df[['arrival_count']].resample('ME').sum()

def assign_period(date):
    if date < pd.Timestamp('2000-01-01'):
        return 'Legacy (< 2000)'
    elif date < COVID_START:
        return 'Pre-Pandemic (2000-Mar 2020)'
    elif date <= COVID_END:
        return 'Pandemic (Apr 2020-2021)'
    else:
        return 'Post-Pandemic (2022+)'

monthly_df['period'] = monthly_df.index.map(assign_period)
monthly_df.head()
```

```
Out[60]:
```

	arrival_count	period
1989-01-31	291566.0	Legacy (< 2000)
1989-02-28	197348.0	Legacy (< 2000)
1989-03-31	149001.0	Legacy (< 2000)
1989-04-30	84867.0	Legacy (< 2000)
1989-05-31	69293.0	Legacy (< 2000)

```
In [61]: period_colors = {
    'Legacy (< 2000)': '#9e9e9e',
    'Pre-Pandemic (2000-Mar 2020)': '#2196f3',
    'Pandemic (Apr 2020-2021)': '#f44336',
    'Post-Pandemic (2022+)': '#4caf50',
}

fig, ax = plt.subplots(figsize=(14, 5))

for period, color in period_colors.items():
    segment = monthly_df[monthly_df['period'] == period]
    if segment.empty:
        continue
    ax.plot(
        segment.index,
        segment['arrival_count'],
        color=color,
```

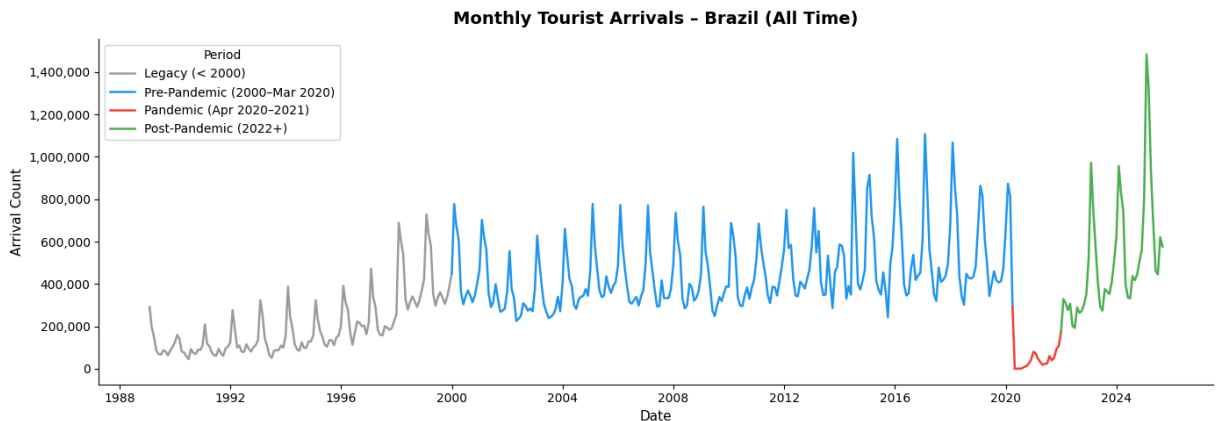
```

        linewidth=1.8,
        label=period,
    )

    # stitch consecutive segments so there are no visual gaps at boundaries
    for i in range(len(monthly_df) - 1):
        curr, nxt = monthly_df.iloc[i], monthly_df.iloc[i + 1]
        if curr['period'] != nxt['period']:
            ax.plot(
                [monthly_df.index[i], monthly_df.index[i + 1]],
                [curr['arrival_count'], nxt['arrival_count']],
                color=period_colors[nxt['period']],
                linewidth=1.8,
            )

    ax.set_title(
        'Monthly Tourist Arrivals - Brazil (All Time)',
        fontsize=14,
        fontweight='bold',
        pad=12,
    )
    ax.set_xlabel('Date', fontsize=11)
    ax.set_ylabel('Arrival Count', fontsize=11)
    ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
    ax.legend(title='Period', frameon=True)
    sns.despine()
    plt.tight_layout()

```



```

In [62]: period_order = [
    'Legacy (< 2000)',
    'Pre-Pandemic (2000-Mar 2020)',
    'Pandemic (Apr 2020-2021)',
    'Post-Pandemic (2022+)',
]
period_colors_list = [period_colors[p] for p in period_order]

period_stats = (
    monthly_df
    .groupby('period')['arrival_count']
    .agg(
        mean='mean',
        median='median',

```

```

        cv=lambda x: x.std() / x.mean() * 100,
        range=lambda x: x.max() - x.min(),
    )
    .reindex(period_order)
)

metrics = [
    ('mean', 'Mean Arrivals', lambda v: f'{int(v):,}'),
    ('median', 'Median Arrivals', lambda v: f'{int(v):,}'),
    ('cv', 'Coefficient of Variation (%)', lambda v: f'{v:.1f}%'),
    ('range', 'Range (Max - Min)', lambda v: f'{int(v):,}'),
]

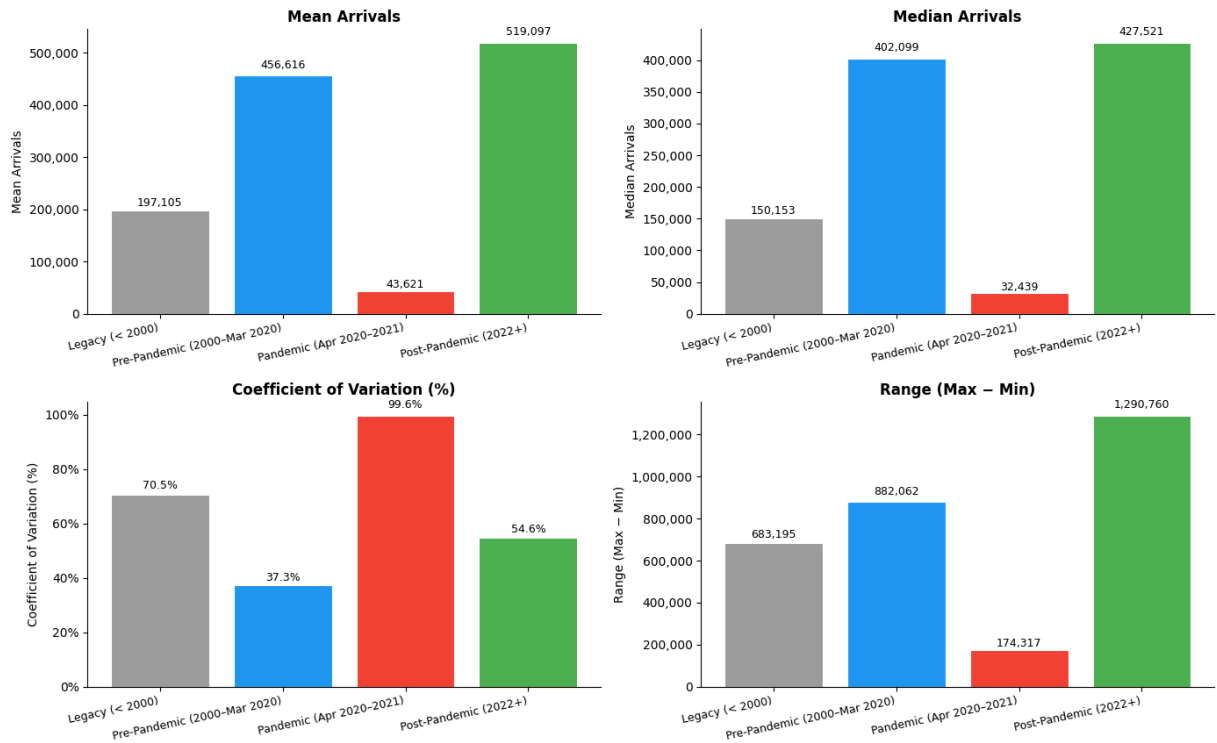
fig, axes = plt.subplots(2, 2, figsize=(14, 9))
fig.suptitle('Arrival Statistics by Period', fontsize=15, fontweight='bold', y=1.01)

for ax, (col, title, fmt) in zip(axes.flat, metrics):
    bars = ax.bar(
        range(len(period_order)),
        period_stats[col],
        color=period_colors_list,
        edgecolor='white',
        linewidth=0.8,
    )
    for bar, val in zip(bars, period_stats[col]):
        ax.text(
            bar.get_x() + bar.get_width() / 2,
            bar.get_height() * 1.02,
            fmt(val),
            ha='center',
            va='bottom',
            fontsize=9,
        )
    ax.set_title(title, fontsize=12, fontweight='bold')
    ax.set_ylabel(title, fontsize=10)
    ax.set_xticks(range(len(period_order)))
    ax.set_xticklabels(period_order, rotation=12, ha='right', fontsize=9)
    is_cv = col == 'cv'
    ax.yaxis.set_major_formatter(
        plt.FuncFormatter(
            lambda x, _, _is_cv=is_cv: f'{x:.0f}%' if _is_cv else f'{x:,.0f}'
        )
    )
    sns.despine(ax=ax)

plt.tight_layout()
plt.show()

```

Arrival Statistics by Period

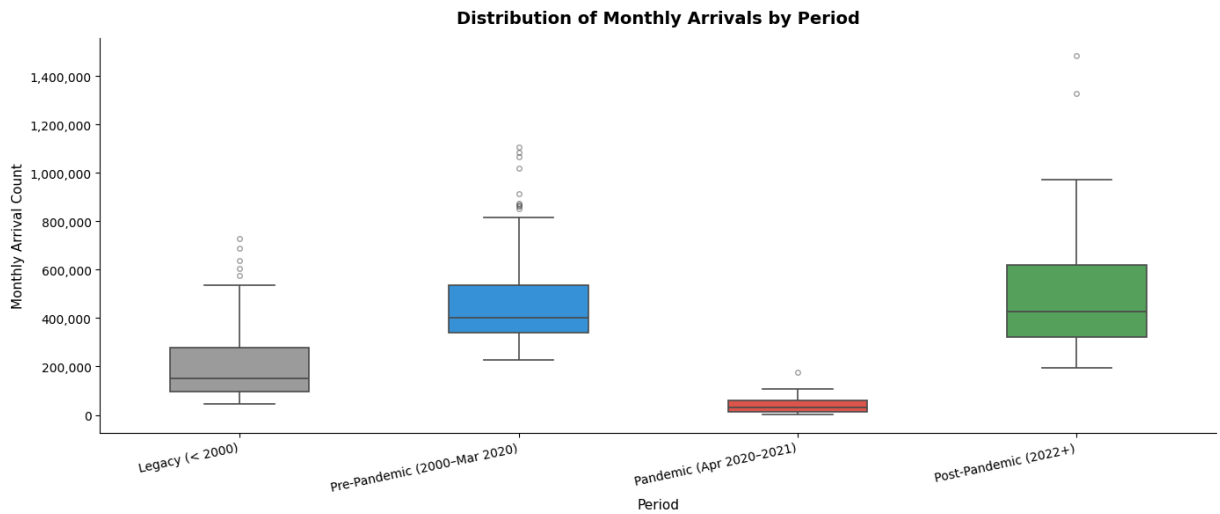


```
In [63]: fig, ax = plt.subplots(figsize=(14, 6))

sns.boxplot(
    data=monthly_df,
    x='period',
    y='arrival_count',
    hue='period',
    order=period_order,
    palette=period_colors,
    width=0.5,
    linewidth=1.2,
    flierprops=dict(marker='o', markersize=4, alpha=0.5),
    legend=False,
    ax=ax,
)

ax.set_title(
    'Distribution of Monthly Arrivals by Period', fontsize=14, fontweight='bold', p
)

ax.set_xlabel('Period', fontsize=11)
ax.set_ylabel('Monthly Arrival Count', fontsize=11)
ax.set_xticks(range(len(period_order)))
ax.set_xticklabels(period_order, rotation=12, ha='right', fontsize=10)
ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
sns.despine()
plt.tight_layout()
plt.show()
```



Trend

```
In [64]: import matplotlib.patches as mpatches

trend_periods = [
    (
        'Pre-Pandemic (2000-Mar 2020)',
        pd.Timestamp('2000-01-01'),
        COVID_START - pd.Timedelta(days=1),
    ),
    ('Pandemic (Apr 2020-2021)', COVID_START, COVID_END),
    ('Post-Pandemic (2022+)', pd.Timestamp('2022-01-01'), monthly_df.index[-1]),
]

post_2000 = monthly_df[monthly_df['period'] != 'Legacy (< 2000)']

fig = plt.figure(figsize=(16, 10))
gs = fig.add_gridspec(2, 3, hspace=0.5, wspace=0.3)

# — top: full post-2000 spanning all 3 columns —————
ax_full = fig.add_subplot(gs[0, :])

ax_full.plot(
    post_2000.index,
    post_2000['arrival_count'],
    color='#555555',
    linewidth=1.2,
    label='Monthly Arrivals',
)

rolling_full = post_2000['arrival_count'].rolling(12, center=True).mean()
ax_full.plot(
    post_2000.index,
    rolling_full,
    color='orange',
    linewidth=2,
    linestyle='--',
    label='12-month Rolling Mean',
)
```

```

for period, start, end in trend_periods:
    ax_full.axvspan(start, end, alpha=0.12, color=period_colors[period])

line_handles, _ = ax_full.get_legend_handles_labels()
span_handles = [
    mpatches.Patch(color=period_colors[p], alpha=0.4, label=p)
    for p, _, _ in trend_periods
]
ax_full.legend(handles=line_handles + span_handles, fontsize=9, frameon=True)
ax_full.set_title(
    'Visual Trend Inspection - Full Period (2000-present)',
    fontsize=13,
    fontweight='bold',
)
ax_full.set_xlabel('Date', fontsize=10)
ax_full.set_ylabel('Arrival Count', fontsize=10)
ax_full.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
sns.despine(ax=ax_full)

# — bottom: one subplot per period —————
for col_idx, (period, start, end) in enumerate(trend_periods):
    ax = fig.add_subplot(gs[1, col_idx])
    color = period_colors[period]

    segment = monthly_df[(monthly_df.index >= start) & (monthly_df.index <= end)]

    ax.plot(
        segment.index,
        segment['arrival_count'],
        color=color,
        linewidth=1.4,
        label='Monthly Arrivals',
    )
    ax.axvspan(start, end, alpha=0.1, color=color)

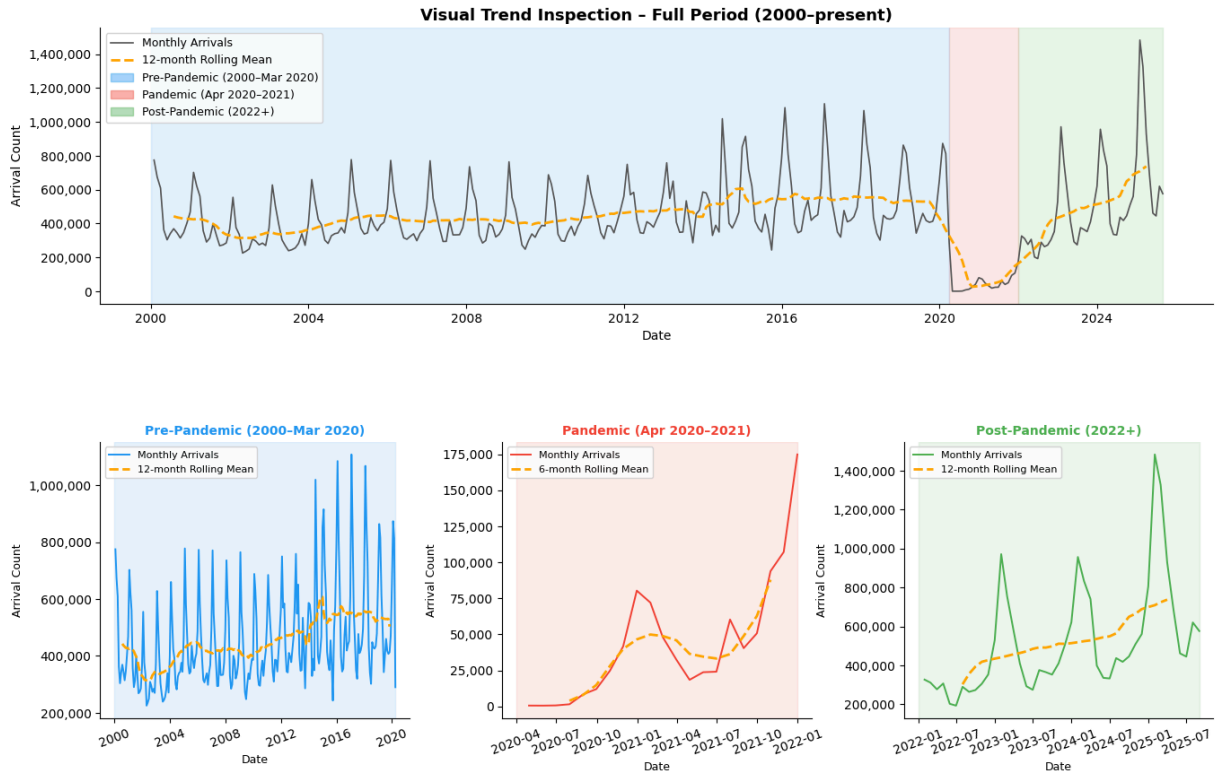
    # pandemic (~24 obs) is too short for a 12-month centered window
    window = 6 if len(segment) < 36 else 12
    rolling = segment['arrival_count'].rolling(window, center=True).mean()
    ax.plot(
        segment.index,
        rolling,
        color='orange',
        linewidth=2,
        linestyle='--',
        label=f'{window}-month Rolling Mean',
    )

    ax.set_title(period, fontsize=10, fontweight='bold', color=color)
    ax.set_xlabel('Date', fontsize=9)
    ax.set_ylabel('Arrival Count', fontsize=9)
    ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
    ax.legend(fontsize=8, frameon=True)
    ax.tick_params(axis='x', rotation=20)
    sns.despine(ax=ax)

```

```
plt.suptitle(
    'Visual Trend Inspection by Period', fontsize=14, fontweight='bold', y=1.01
)
plt.show()
```

Visual Trend Inspection by Period



```
In [65]: def fit_trends(series):
    """Return (lin_model, poly_model, poly_features, t_index, stats_dict)."""
    t = np.arange(len(series)).reshape(-1, 1)
    y = series.values

    lin = LinearRegression().fit(t, y)

    pf = PolynomialFeatures(degree=2)
    poly = LinearRegression().fit(pf.fit_transform(t), y)

    stats = {
        'slope': lin.coef_[0],
        'r2_lin': lin.score(t, y),
        'r2_poly': poly.score(pf.fit_transform(t), y),
    }
    return lin, poly, pf, t, stats

def plot_trend_panel(ax, segment, period, color):
    lin, poly, pf, t, stats = fit_trends(segment['arrival_count'])

    ax.plot(
        segment.index,
        segment['arrival_count'],
```

```

        color=color,
        linewidth=1.2,
        alpha=0.6,
        label='Monthly Arrivals',
    )
    ax.plot(
        segment.index, lin.predict(t), color='red', linewidth=2, label='Linear Tren
    )
    ax.plot(
        segment.index,
        poly.predict(pf.fit_transform(t)),
        color='purple',
        linewidth=2,
        linestyle='--',
        label='Polynomial Trend',
    )

    ax.annotate(
        f'slope: {stats["slope"]:+.0f}/mo\nR² lin: {stats["r2_lin"]:.3f}\nR² poly:
        xy=(0.03, 0.95),
        xycoords='axes fraction',
        va='top',
        fontsize=8,
        bbox=dict(boxstyle='round,pad=0.3', fc='white', alpha=0.7),
    )
    ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
    ax.legend(fontsize=8, frameon=True)
    sns.despine(ax=ax)
    return stats

```

— build figure

```
post_2000 = monthly_df[monthly_df['period'] != 'Legacy (< 2000)']
```

```
fig = plt.figure(figsize=(16, 10))
```

```
gs = fig.add_gridspec(2, 3, hspace=0.5, wspace=0.3)
```

top: full post-2000

```
ax_full = fig.add_subplot(gs[0, :])
```

```
full_stats = plot_trend_panel(ax_full, post_2000, 'Full Period', '#555555')
```

```
for period, start, end in trend_periods:
```

```
    ax_full.axvspan(start, end, alpha=0.1, color=period_colors[period])
```

```
ax_full.set_title(
```

```
    'Trend Estimation - Full Period (2000-present)', fontsize=13, fontweight='bold'
```

```
)
```

```
ax_full.set_xlabel('Date', fontsize=10)
```

```
ax_full.set_ylabel('Arrival Count', fontsize=10)
```

bottom: one per period

```
all_stats = [('Full Period (2000-present)', full_stats)]
```

```
for col_idx, (period, start, end) in enumerate(trend_periods):
```

```
    ax = fig.add_subplot(gs[1, col_idx])
```

```
    color = period_colors[period]
```

```

segment = monthly_df[(monthly_df.index >= start) & (monthly_df.index <= end)]
stats = plot_trend_panel(ax, segment, period, color)
ax.axvspan(start, end, alpha=0.08, color=color)
ax.set_title(period, fontsize=10, fontweight='bold', color=color)
ax.set_xlabel('Date', fontsize=9)
ax.set_ylabel('Arrival Count', fontsize=9)
ax.tick_params(axis='x', rotation=20)

all_stats.append((period, stats))

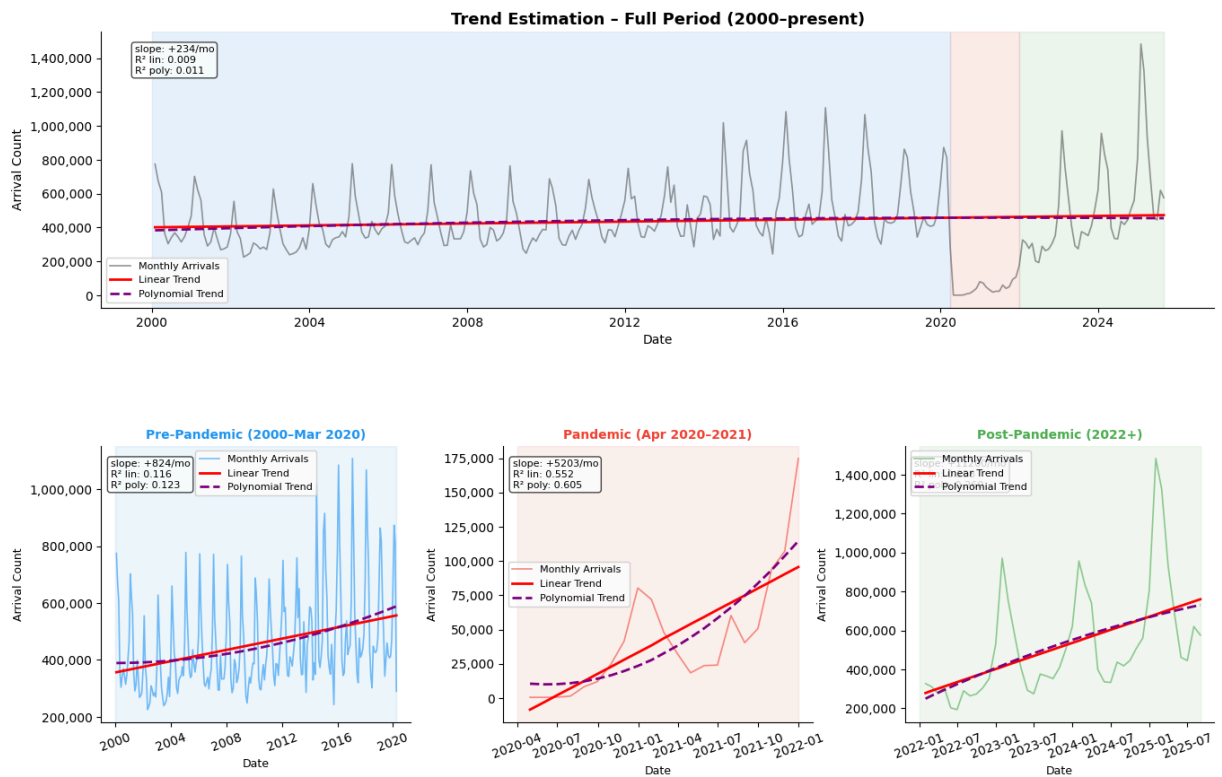
plt.suptitle(
    'Linear & Polynomial Trend Estimation by Period',
    fontsize=14,
    fontweight='bold',
    y=1.01,
)
plt.show()

# — summary table —————
summary = pd.DataFrame(
    [
        (p, f'{s["slope"]:+.1f}', f'{s["r2_lin"]:.4f}', f'{s["r2_poly"]:.4f}')
        for p, s in all_stats
    ],
    columns=['Period', 'Linear Slope (arrivals/month)', 'R² Linear', 'R² Polynomial']
).set_index('Period')

print(summary.to_string())

```

Linear & Polynomial Trend Estimation by Period



Period	Linear Slope (arrivals/month)	R ² Linear	R ² Polynomial
Full Period (2000–present)	+233.5	0.0094	0.0108
Pre-Pandemic (2000–Mar 2020)	+823.9	0.1156	0.1230
Pandemic (Apr 2020–2021)	+5202.9	0.5520	0.6050
Post-Pandemic (2022+)	+11199.7	0.2576	0.2601

Statistical Trend Tests

The visual and regression checks above are complemented with the non-parametric tests wrapped by `check_trend` and `check_deterministic_trend` (Joseph & Tackes, *Modern Time Series Forecasting with Python* 2E, `src/transforms/stationary_utils.py`):

- **Kendall's τ** : rank correlation between the series and time.
- **Mann-Kendall**: monotonic-trend test with Sen's slope. For series with fewer than 50 points the helper applies the pre-whitening variant to reduce the effect of autocorrelation.
- **Seasonal Mann-Kendall ($m = 12$)**: runs the test within each calendar month, so the seasonal cycle does not hide or inflate the trend. The slope is in arrivals per year.
- **Deterministic trend**: ADF with constant only (`c`) vs constant + trend (`ct`). If the series is non-stationary under `c` but stationary under `ct`, the trend is deterministic and detrending is enough. Otherwise differencing is needed.

```
In [66]: # Helpers `check_trend` / `check_deterministic_trend` from Joseph & Tackes,
# "Modern Time Series Forecasting with Python" 2E – src/transforms/stationary_utils

period_series = {
    'Full Period (2000–present)': post_2000['arrival_count'],
    **{
        period: monthly_df.loc[
            (monthly_df.index >= start) & (monthly_df.index <= end), 'arrival_count'
        ]
        for period, start, end in trend_periods
    },
}

period_series_colors = {
    'Full Period (2000–present)': '#555555',
    **{p: period_colors[p] for p, _, _ in trend_periods},
}

trend_rows = []
for label, series in period_series.items():
    with warnings.catch_warnings():
        warnings.simplefilter('ignore')
        kt = check_trend(series, confidence=0.05)
        mk_res = check_trend(series, confidence=0.05, mann_kendall=True)
        smk = check_trend(
            series, confidence=0.05, mann_kendall=True, seasonal_period=SEASONAL_PERIOD
        )
        det = check_deterministic_trend(series, confidence=0.05)

    trend_rows.append({
```

```

        'Period': label,
        'n': len(series),
        'Kendall  $\tau$ ': f'{kt.slope:+.3f}',
        'Kendall p': f'{kt.p_value:.4f}',
        'Kendall trend': '✓' if kt.trend else 'X',
        'MK test': type(mk_res).__name__.replace('_', ' '),
        "MK Sen's slope (/mo)": f'{mk_res.slope:+,.0f}',
        'MK p': f'{mk_res.p_value:.4f}',
        'MK trend': '✓' if mk_res.trend else 'X',
        'Seasonal MK slope (/yr)': f'{smk.slope:+,.0f}',
        'Seasonal MK p': f'{smk.p_value:.4f}',
        'Seasonal MK trend': '✓' if smk.trend else 'X',
        'ADF(c) p': f'{det.adf_res.results[1]:.4f}',
        'ADF(ct) p': f'{det.adf_ct_res.results[1]:.4f}',
        'Deterministic trend': '✓' if det.deterministic_trend else 'X',
    })

```

```

trend_tests = pd.DataFrame(trend_rows).set_index('Period')

```

```

print('=' * 100)
print('TREND TESTS – Kendall  $\tau$ , Mann-Kendall, Seasonal MK (m=12), ADF c vs ct')
print('H0 (Kendall / MK): no monotonic trend → reject (p < 0.05) = trend present')
print('=' * 100)
print(trend_tests.T.to_string())

```

```

=====
TREND TESTS – Kendall  $\tau$ , Mann-Kendall, Seasonal MK (m=12), ADF c vs ct
H0 (Kendall / MK): no monotonic trend  $\rightarrow$  reject ( $p < 0.05$ ) = trend present
=====

```

Period	Full Period (2000-present)	Pre-Pandemic (2000-Mar 2020)	Post-Pandemic (2022+)
Pandemic (Apr 2020-2021)			
n	308		243
21	44		
Kendall τ	+0.101		+0.265
+0.657	+0.455		
Kendall p	0.0084		0.0000
0.0000	0.0000		
Kendall trend	✓		✓
✓	✓		
MK test	Mann Kendall Test		Mann Kendall Test
Modified Mann Kendall Test	PreWhitening Approach		Modified Mann Kendall Test
PreWhitening Approach			PreWhitening Approach
MK Sen's slope (/mo)	+266		+661
+4,655	+8,162		
MK p	0.0084		0.0000
0.0058	0.1070		
MK trend	✓		✓
✓	X		
Seasonal MK slope (/yr)	+5,399		+8,062
+38,665	+104,210		
Seasonal MK p	0.0000		0.0000
0.0077	0.0000		
Seasonal MK trend	✓		✓
✓	✓		
ADF(c) p	0.1356		0.6776
0.9991	0.8104		
ADF(ct) p	0.3341		0.1252
0.0002	0.0002		
Deterministic trend	X		X
✓	✓		

```

In [67]: def additive_or_multiplicative(segment, label, color):
          """Visual + quantitative check for decomposition type."""
          series = segment['arrival_count'].dropna()

          annual = series.resample('YE').agg(['mean', 'std']).dropna()
          corr = annual['mean'].corr(annual['std'])

          if corr > 0.7:
              verdict = 'Multiplicative (corr > 0.7)'
              verdict_color = '#e53935'
          elif corr < 0.4:
              verdict = 'Additive (corr < 0.4)'
              verdict_color = '#1e88e5'
          else:
              verdict = 'Ambiguous (0.4 ≤ corr ≤ 0.7)'
              verdict_color = '#fb8c00'

          return {

```

```

        'label': label,
        'corr': corr,
        'verdict': verdict,
        'verdict_color': verdict_color,
        'series': series,
        'annual': annual,
        'color': color,
    }

non_legacy_periods = [
    (
        'Pre-Pandemic (2000-Mar 2020)',
        pd.Timestamp('2000-01-01'),
        COVID_START - pd.Timedelta(days=1),
    ),
    ('Pandemic (Apr 2020-2021)', COVID_START, COVID_END),
    ('Post-Pandemic (2022+)', pd.Timestamp('2022-01-01'), monthly_df.index[-1]),
]

post_2000 = monthly_df[monthly_df['period'] != 'Legacy (< 2000)']

results = [
    additive_or_multiplicative(post_2000, 'Full Period (2000-present)', '#555555')
]
for period, start, end in non_legacy_periods:
    seg = monthly_df[(monthly_df.index >= start) & (monthly_df.index <= end)]
    results.append(additive_or_multiplicative(seg, period, period_colors[period]))

# — 4 × 2 figure: raw series vs annual mean-std relationship —————
fig, axes = plt.subplots(4, 2, figsize=(15, 14), constrained_layout=True)
fig.suptitle(
    'Additive vs Multiplicative Check\n(Raw Series vs Annual Mean-Std Relationship)'
    fontsize=14,
    fontweight='bold',
)

for row, res in enumerate(results):
    ax_raw, ax_ms = axes[row, 0], axes[row, 1]
    color = res['color']

    ax_raw.plot(res['series'].index, res['series'], color=color, linewidth=1.2)
    ax_raw.set_title(
        f'{res["label"]} - Raw', fontsize=10, fontweight='bold', color=color
    )
    ax_raw.set_ylabel('Arrival Count', fontsize=9)
    ax_raw.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
    sns.despine(ax=ax_raw)

    annual = res['annual']
    ax_ms.scatter(annual['mean'], annual['std'], color=color, s=35, alpha=0.8)
    if len(annual) > 1:
        coef = np.polyfit(annual['mean'], annual['std'], 1)
        x_line = np.linspace(annual['mean'].min(), annual['mean'].max(), 50)
        ax_ms.plot(x_line, np.polyval(coef, x_line), color=color, linestyle='--')
    ax_ms.set_title(

```

```

        f'{res["label"]} - Annual Mean vs Std',
        fontsize=10,
        fontweight='bold',
        color=color,
    )
    ax_ms.set_xlabel('Annual Mean', fontsize=9)
    ax_ms.set_ylabel('Annual Std', fontsize=9)
    ax_ms.xaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
    ax_ms.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
    ax_ms.annotate(
        f'mean-std corr: {res["corr"]:.3f}\n→ {res["verdict"]}',
        xy=(0.03, 0.95),
        xycoords='axes fraction',
        va='top',
        fontsize=8.5,
        color=res['verdict_color'],
        bbox=dict(boxstyle='round,pad=0.3', fc='white', alpha=0.8),
    )
    sns.despine(ax=ax_ms)

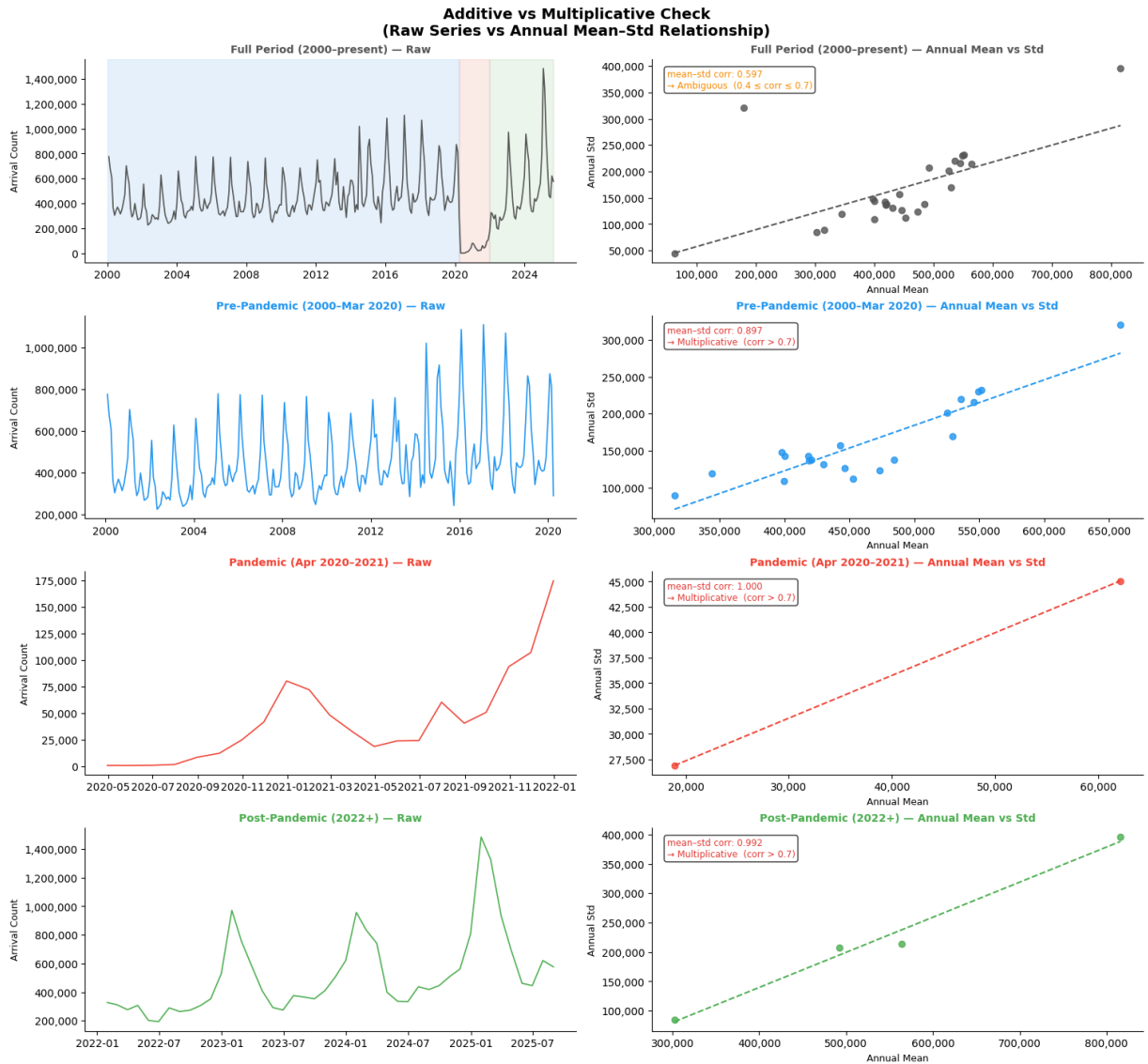
    # period shading on full-period row only
    if res['label'] == 'Full Period (2000-present)':
        for period, start, end in non_legacy_periods:
            ax_raw.axvspan(start, end, alpha=0.1, color=period_colors[period])

plt.show()

# — summary table —————
summary_add = pd.DataFrame(
    [(r['label'], f'{r["corr"]:.3f}', r['verdict']) for r in results],
    columns=['Period', 'Mean-Std Correlation', 'Verdict'],
).set_index('Period')

print(summary_add.to_string())

```



Period	Mean-Std Correlation	Verdict
Full Period (2000-present)	0.597	Ambiguous ($0.4 \leq \text{corr} \leq 0.7$)
Pre-Pandemic (2000-Mar 2020)	0.897	Multiplicative ($\text{corr} > 0.7$)
Pandemic (Apr 2020-2021)	1.000	Multiplicative ($\text{corr} > 0.7$)
Post-Pandemic (2022+)	0.992	Multiplicative ($\text{corr} > 0.7$)

Seasonal

```
In [68]: stl_series = monthly_df.loc[monthly_df['period'] != 'Legacy (< 2000)', 'arrival_cou

stl = STL(stl_series, period=12, seasonal=13, robust=True)
result = stl.fit()

components = [
    ('observed', 'Observed', result.observed),
    ('trend', 'Trend', result.trend),
    ('seasonal', 'Seasonal', result.seasonal),
    ('resid', 'Residual', result.resid),
]

period_spans = [
```

```

(
    'Pre-Pandemic (2000-Mar 2020)',
    pd.Timestamp('2000-01-01'),
    COVID_START - pd.Timedelta(days=1),
),
('Pandemic (Apr 2020-2021)', COVID_START, COVID_END),
('Post-Pandemic (2022+)', pd.Timestamp('2022-01-01'), stl_series.index[-1]),
]

fig, axes = plt.subplots(4, 1, figsize=(14, 12), sharex=True)
fig.suptitle(
    'STL Seasonal Decomposition (2000-present)',
    fontsize=14,
    fontweight='bold',
    y=1.01,
)

for ax, (_, label, component) in zip(axes, components):
    ax.plot(component.index, component.values, color='#333333', linewidth=1.2)

    for period, start, end in period_spans:
        ax.axvspan(start, end, alpha=0.12, color=period_colors[period], label=period)

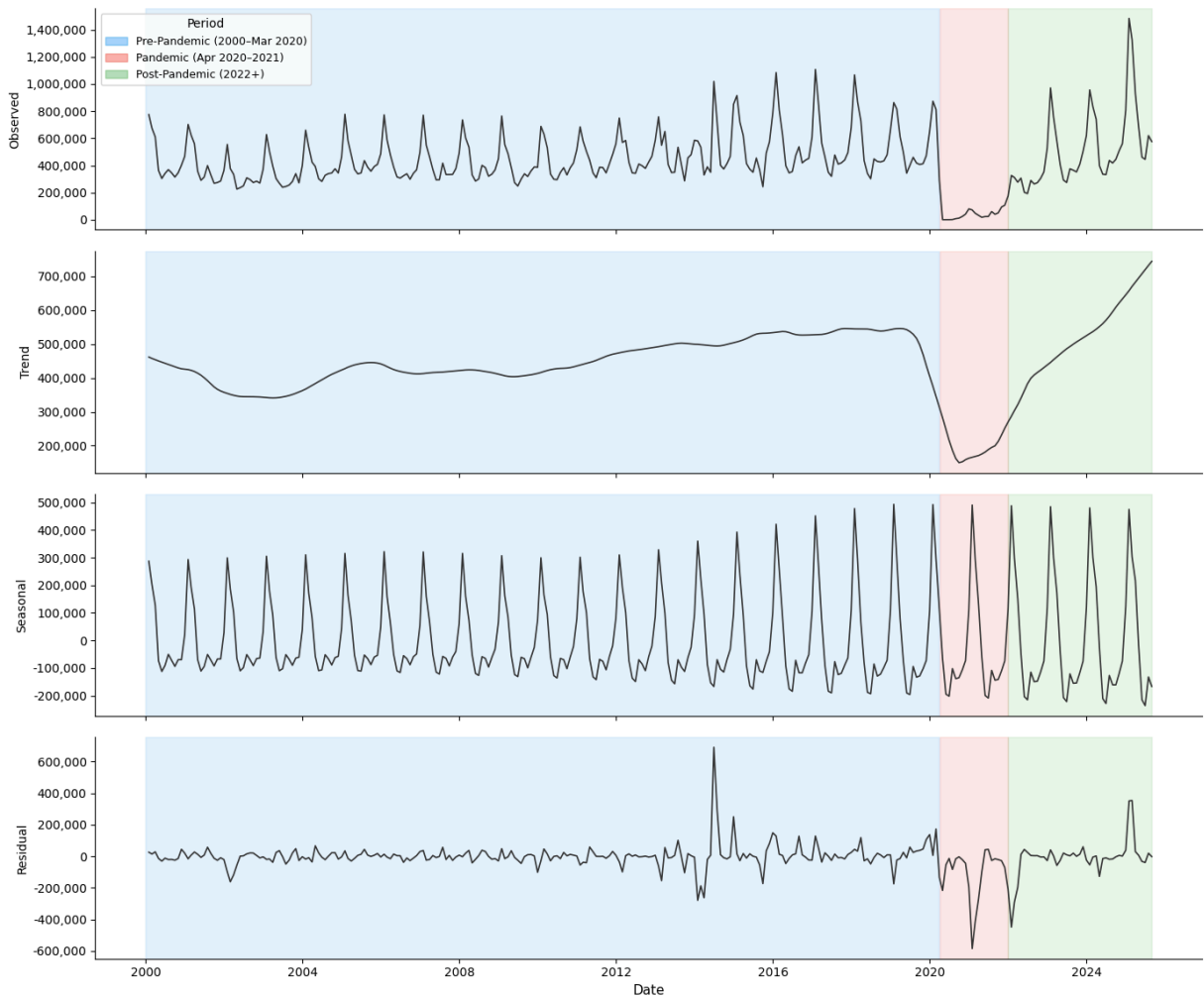
    ax.set_ylabel(label, fontsize=10)
    ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
    sns.despine(ax=ax)

# single shared legend on the top subplot
handles = [
    plt.Rectangle((0, 0), 1, 1, color=period_colors[p], alpha=0.4)
    for p in [
        'Pre-Pandemic (2000-Mar 2020)',
        'Pandemic (Apr 2020-2021)',
        'Post-Pandemic (2022+)',
    ]
]
labels = [
    'Pre-Pandemic (2000-Mar 2020)',
    'Pandemic (Apr 2020-2021)',
    'Post-Pandemic (2022+)',
]
axes[0].legend(
    handles, labels, title='Period', loc='upper left', frameon=True, fontsize=9
)

axes[-1].set_xlabel('Date', fontsize=11)
plt.tight_layout()
plt.show()

```

STL Seasonal Decomposition (2000-present)



```
In [69]: def stl_seasonal_strength(result):
    """max(0, 1 - Var(resid) / Var(seasonal + resid))"""
    var_resid = np.var(result.resid)
    var_seasonal_resid = np.var(result.seasonal + result.resid)
    return max(0.0, 1 - var_resid / var_seasonal_resid)

seasonal_periods = [
    (
        'Pre-Pandemic (2000-Mar 2020)',
        pd.Timestamp('2000-01-01'),
        COVID_START - pd.Timedelta(days=1),
    ),
    ('Pandemic (Apr 2020-2021)', COVID_START, COVID_END),
    ('Post-Pandemic (2022+)', pd.Timestamp('2022-01-01'), monthly_df.index[-1]),
]

# — fit STL for full period + each sub-period —————
post_2000 = monthly_df[monthly_df['period'] != 'Legacy (< 2000)']

full_series = post_2000['arrival_count'].dropna()
full_result = STL(full_series, period=12, seasonal=13, robust=True).fit()
full_strength = stl_seasonal_strength(full_result)
```

```

period_stl = {}
for period, start, end in seasonal_periods:
    seg = monthly_df[(monthly_df.index >= start) & (monthly_df.index <= end)]
    series = seg['arrival_count'].dropna()
    # pandemic is too short for STL with seasonal=13; use seasonal=7
    s_window = 7 if len(series) < 36 else 13
    result = STL(series, period=12, seasonal=s_window, robust=True).fit()
    period_stl[period] = dict(result=result, strength=stl_seasonal_strength(result))

# — 1x3 seasonal component plot —————
fig, axes = plt.subplots(1, 3, figsize=(16, 4), constrained_layout=True)
fig.suptitle('STL Seasonal Component by Period', fontsize=14, fontweight='bold')

for ax, (period, start, end) in zip(axes, seasonal_periods):
    color = period_colors[period]
    res = period_stl[period]['result']
    fs = period_stl[period]['strength']

    ax.plot(res.seasonal.index, res.seasonal.values, color=color, linewidth=1.4)
    ax.axhline(0, color='black', linewidth=0.7, linestyle='--')
    ax.axvspan(start, end, alpha=0.08, color=color)

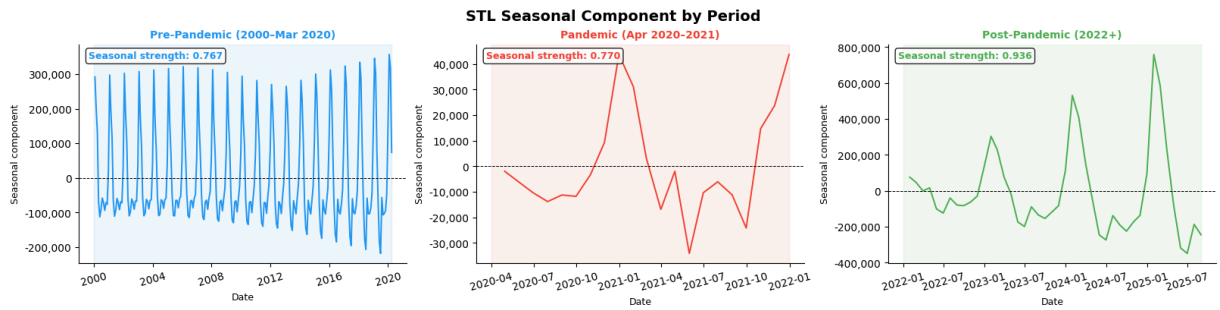
    ax.annotate(
        f'Seasonal strength: {fs:.3f}',
        xy=(0.03, 0.97),
        xycoords='axes fraction',
        va='top',
        fontsize=9,
        color=color,
        fontweight='bold',
        bbox=dict(boxstyle='round,pad=0.3', fc='white', alpha=0.85),
    )
    ax.set_title(period, fontsize=10, fontweight='bold', color=color)
    ax.set_xlabel('Date', fontsize=9)
    ax.set_ylabel('Seasonal component', fontsize=9)
    ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
    ax.tick_params(axis='x', rotation=15)
    sns.despine(ax=ax)

plt.show()

# — summary table —————
summary_seasonal = pd.DataFrame(
    [
        ('Full Period (2000-present)', f'{full_strength:.4f}')
        + [(p, f'{period_stl[p]["strength"]:.4f}') for p, _, _ in seasonal_periods],
        columns=['Period', 'Seasonal Strength'],
    ).set_index('Period')

print(summary_seasonal.to_string())

```



Seasonal Strength

Period	
Full Period (2000-present)	0.7159
Pre-Pandemic (2000-Mar 2020)	0.7674
Pandemic (Apr 2020-2021)	0.7700
Post-Pandemic (2022+)	0.9355

```
In [70]: month_labels = [
    'Jan',
    'Feb',
    'Mar',
    'Apr',
    'May',
    'Jun',
    'Jul',
    'Aug',
    'Sep',
    'Oct',
    'Nov',
    'Dec',
]

shape_panels = [
    (
        'Full Period (2000-present)',
        monthly_df[monthly_df['period'] != 'Legacy (< 2000)'],
        '#555555',
    ),
    (
        'Pre-Pandemic (2000-Mar 2020)',
        monthly_df[monthly_df['period'] == 'Pre-Pandemic (2000-Mar 2020)'],
        period_colors['Pre-Pandemic (2000-Mar 2020)'],
    ),
    (
        'Pandemic (Apr 2020-2021)',
        monthly_df[monthly_df['period'] == 'Pandemic (Apr 2020-2021)'],
        period_colors['Pandemic (Apr 2020-2021)'],
    ),
    (
        'Post-Pandemic (2022+)',
        monthly_df[monthly_df['period'] == 'Post-Pandemic (2022+)'],
        period_colors['Post-Pandemic (2022+)'],
    ),
]

fig, axes = plt.subplots(2, 2, figsize=(14, 9), constrained_layout=True)
fig.suptitle(
```

```

    'Seasonal Shape: Mean Arrivals by Month per Period', fontsize=14, fontweight='b
)

for ax, (label, segment, color) in zip(axes.flat, shape_panels):
    monthly_means = (
        segment['arrival_count']
        .groupby(segment.index.month)
        .mean()
        .reindex(range(1, 13))
    )

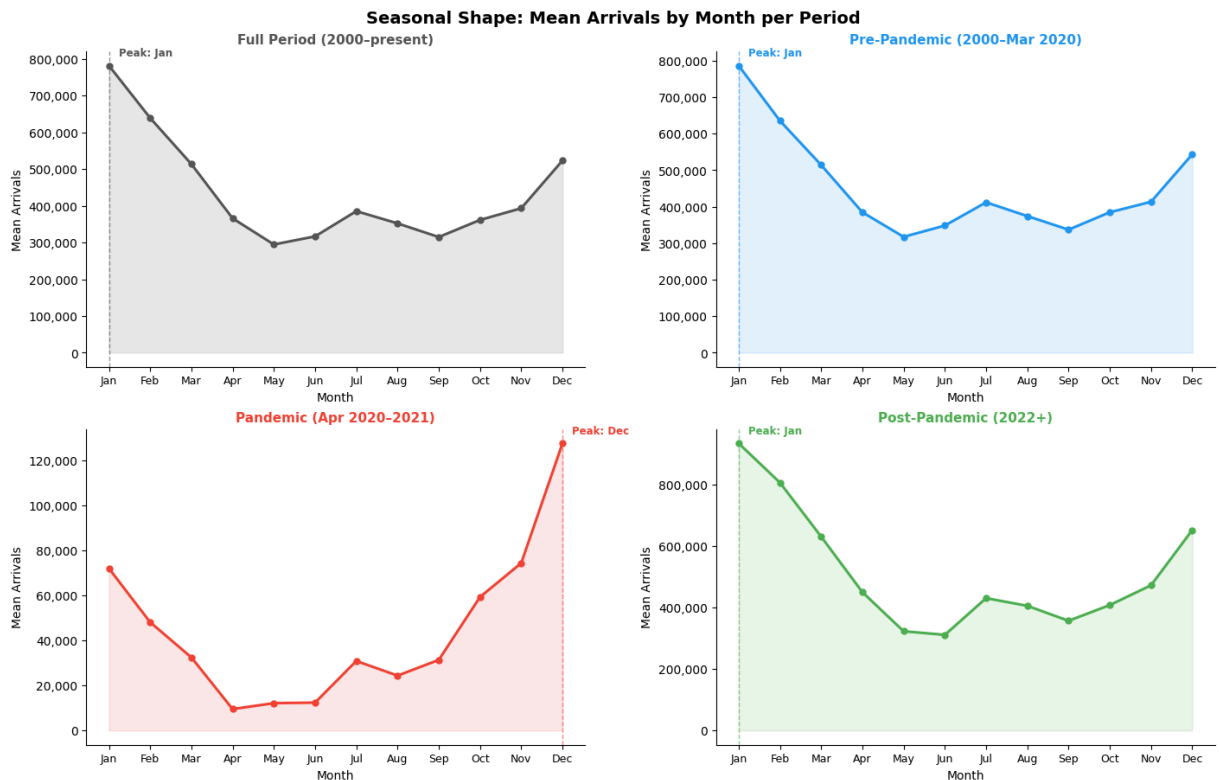
    ax.plot(
        range(1, 13),
        monthly_means.values,
        color=color,
        linewidth=2.2,
        marker='o',
        markersize=5,
    )
    ax.fill_between(range(1, 13), monthly_means.values, alpha=0.12, color=color)

    peak_month = monthly_means.idxmax()
    ax.axvline(peak_month, color=color, linewidth=1, linestyle='--', alpha=0.6)
    ax.annotate(
        f'Peak: {month_labels[peak_month - 1]}',
        xy=(peak_month, monthly_means[peak_month]),
        xytext=(8, 8),
        textcoords='offset points',
        fontsize=8.5,
        color=color,
        fontweight='bold',
    )

    ax.set_title(label, fontsize=11, fontweight='bold', color=color)
    ax.set_xlabel('Month', fontsize=10)
    ax.set_ylabel('Mean Arrivals', fontsize=10)
    ax.set_xticks(range(1, 13))
    ax.set_xticklabels(month_labels, fontsize=9)
    ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{int(x):,}'))
    sns.despine(ax=ax)

plt.show()

```



Statistical Seasonality Test

`check_seasonality` (Joseph & Tackes, *Modern Time Series Forecasting with Python* 2E, `src/transforms/stationary_utils.py`, adapted by the authors from `darts`) finds local maxima of the ACF and accepts a candidate lag m when its autocorrelation exceeds a Bartlett-formula confidence band. It is run in two modes:

- **Inferred:** the first significant ACF peak, searched up to `max_lag`.
- **Tested at $m = 12$:** checks whether lag 12 is a local ACF maximum and significant.

A strong trend dominates the ACF and can hide the seasonal peaks, so the test is also repeated after linear detrending with `DetrendingTransformer` (same source, `src/transforms/target_transformations.py`), which is the order the `AutoStationaryTransformer` uses.

```
In [71]: # Helpers `check_seasonality` / `DetrendingTransformer` from Joseph & Tackes,
# "Modern Time Series Forecasting with Python" 2E - src/transforms/
```

```
def seasonality_row(series, variant):
    n = len(series)
    inferred = check_seasonality(
        series, max_lag=min(36, n - 1), confidence=0.05, verbose=False
    )
    at_12 = (
        check_seasonality(
            series,
            max_lag=min(36, n - 1),
            seasonal_period=SEASONAL_PERIOD,
```

```

        confidence=0.05,
        verbose=False,
    )
    if n - 1 > SEASONAL_PERIOD
    else None
)
return {
    'Variant': variant,
    'n': n,
    'Inferred seasonal': '✓' if inferred.seasonal else 'X',
    'Inferred period': int(inferred.seasonal_periods) if inferred.seasonal else
    'Seasonal at m=12': '-' if at_12 is None else ('✓' if at_12.seasonal else 'X')
}

seasonality_rows = []
for label, series in period_series.items():
    detrended = DetrendingTransformer(degree=1).fit_transform(series, freq=FREQ)
    for variant, s in [('Raw', series), ('Detrended (linear)', detrended)]:
        seasonality_rows.append({'Period': label, **seasonality_row(s, variant)})

seasonality_tests = pd.DataFrame(seasonality_rows).set_index(['Period', 'Variant'])

print('=' * 90)
print('SEASONALITY TEST – ACF local maxima + Bartlett band (check_seasonality)')
print('=' * 90)
print(seasonality_tests.to_string())

```

```

=====
=====
SEASONALITY TEST – ACF local maxima + Bartlett band (check_seasonality)
=====
=====

```

Period	Variant	n	Inferred seasonal	Inferred period
Full Period (2000-present)	Raw	308	✓	
6	✓			
	Detrended (linear)	308	✓	
6	✓			
Pre-Pandemic (2000-Mar 2020)	Raw	243	✓	
12	✓			
	Detrended (linear)	243	✓	
12	✓			
Pandemic (Apr 2020-2021)	Raw	21	✓	
11	X			
	Detrended (linear)	21	✓	
12	✓			
Post-Pandemic (2022+)	Raw	44	✓	
12	✓			
	Detrended (linear)	44	✓	
12	✓			

Stationarity

```

In [72]: def run_adf(series, label):
    """ADF: H0 = unit root (non-stationary). p < 0.05 → stationary."""
    result = adfuller(series.dropna(), autolag='AIC')
    return {
        'label': label,
        'stat': result[0],
        'p': result[1],
        'lags': result[2],
        'stationary_adf': result[1] < 0.05,
    }

def run_kpss(series, label):
    """KPSS: H0 = stationary. p < 0.05 → non-stationary."""
    with warnings.catch_warnings():
        warnings.simplefilter('ignore')
        stat, p, lags, _ = kpss(series.dropna(), regression='c', nlags='auto')
    return {
        'label': label,
        'stat': stat,
        'p': p,
        'lags': lags,
        'stationary_kpss': p >= 0.05,
    }

def joint_verdict(adf_row, kpss_row):
    """
    ADF stationary + KPSS stationary → Stationary
    ADF non-stationary + KPSS non-stationary → Non-stationary
    Disagreement → ambiguous (difference and retest)
    """
    a, k = adf_row['stationary_adf'], kpss_row['stationary_kpss']
    if a and k:
        return 'Stationary'
    if not a and not k:
        return 'Non-stationary'
    return 'Ambiguous'

def test_series(series, label):
    a = run_adf(series, label)
    k = run_kpss(series, label)
    return a, k, joint_verdict(a, k)

def stationarity_grid(periods, transformations):
    """Run ADF + KPSS for every (period, transformation) pair."""
    rows = []
    for period_label, raw_series in periods:
        for trans_label, fn in transformations:
            try:
                s = fn(raw_series, period_label).dropna()
                if len(s) < 8:
                    continue

```

```

        adf_r, kpss_r, verdict = test_series(
            s, f'{period_label} | {trans_label}'
        )
        rows.append({
            'Period': period_label,
            'Transformation': trans_label,
            'ADF stat': round(adf_r['stat'], 4),
            'ADF p': round(adf_r['p'], 4),
            'ADF lags': adf_r['lags'],
            'ADF stationary': '✓' if adf_r['stationary_adf'] else 'X',
            'KPSS stat': round(kpss_r['stat'], 4),
            'KPSS p': round(kpss_r['p'], 4),
            'KPSS stationary': '✓' if kpss_r['stationary_kpss'] else 'X',
            'Joint verdict': verdict,
        })
    except Exception as e:
        rows.append({
            'Period': period_label,
            'Transformation': trans_label,
            'ADF stat': None,
            'ADF p': None,
            'ADF lags': None,
            'ADF stationary': 'ERR',
            'KPSS stat': None,
            'KPSS p': None,
            'KPSS stationary': 'ERR',
            'Joint verdict': str(e),
        })
    return pd.DataFrame(rows)

def plot_stationarity_heatmaps(stat_df, trans_order, period_label_order, title):
    pivot_adf = stat_df.pivot_table(
        index='Period', columns='Transformation', values='ADF p', aggfunc='first'
    ).reindex(index=period_label_order, columns=trans_order)
    pivot_kpss = stat_df.pivot_table(
        index='Period', columns='Transformation', values='KPSS p', aggfunc='first'
    ).reindex(index=period_label_order, columns=trans_order)

    fig, axes = plt.subplots(1, 2, figsize=(18, 4), constrained_layout=True)
    fig.suptitle(
        f'{title}\n(ADF: green = stationary  $p < 0.05$  | KPSS: green = stationary  $p \geq 0.0$ )',
        fontsize=13,
        fontweight='bold',
    )

    # ADF heatmap: Low p → stationary → green
    sns.heatmap(
        pivot_adf.astype(float),
        ax=axes[0],
        annot=True,
        fmt='.3f',
        cmap='RdYlGn_r',
        vmin=0,
        vmax=0.1,
        linewidths=0.4,

```

```

        cbar_kws={'label': 'p-value'},
    )
    axes[0].set_title(
        'ADF p-value (p < 0.05 = stationary)', fontsize=11, fontweight='bold'
    )

    # KPSS heatmap: high p → stationary → green
    sns.heatmap(
        pivot_kpss.astype(float),
        ax=axes[1],
        annot=True,
        fmt='.3f',
        cmap='RdYlGn',
        vmin=0,
        vmax=0.1,
        linewidths=0.4,
        cbar_kws={'label': 'p-value'},
    )
    axes[1].set_title(
        'KPSS p-value (p ≥ 0.05 = stationary)', fontsize=11, fontweight='bold'
    )
    axes[1].set_ylabel('')

    for ax in axes:
        ax.set_xlabel('Transformation', fontsize=9)
        ax.tick_params(axis='x', rotation=30)
        ax.tick_params(axis='y', rotation=0)
    axes[0].set_ylabel('Period', fontsize=9)

    plt.show()

```

— series variants to test —

```

stationarity_periods = [
    (
        'Full (2000-present)',
        monthly_df[monthly_df['period'] != 'Legacy (< 2000)']['arrival_count'],
    ),
    (
        'Pre-Pandemic (2000-Mar 2020)',
        monthly_df[monthly_df['period'] == 'Pre-Pandemic (2000-Mar 2020)']['arrival_count'],
    ),
    (
        'Pandemic (Apr 2020-2021)',
        monthly_df[monthly_df['period'] == 'Pandemic (Apr 2020-2021)']['arrival_count'],
    ),
    (
        'Post-Pandemic (2022+)',
        monthly_df[monthly_df['period'] == 'Post-Pandemic (2022+)']['arrival_count'],
    ),
]
period_label_order = [p for p, _ in stationarity_periods]

```

```

transformations = [
    ('Raw', lambda s, _: s),
    ('Diff(1)', lambda s, _: s.diff(1)),
    ('Diff(12)', lambda s, _: s.diff(12)),
    ('Diff(1)+Diff(12)', lambda s, _: s.diff(1).diff(12)),
]
trans_order = [t for t, _ in transformations]

# — run all tests —————

stat_df = stationarity_grid(stationarity_periods, transformations)

# — display full table —————

display_cols = [
    'Period',
    'Transformation',
    'ADF p',
    'ADF stationary',
    'KPSS p',
    'KPSS stationary',
    'Joint verdict',
]

print('=' * 100)
print('STATIONARITY TEST RESULTS – ADF + KPSS')
print('ADF H0: unit root (non-stationary) → reject (p < 0.05) = stationary')
print('KPSS H0: stationary → reject (p < 0.05) = non-stationary')
print('=' * 100)
print(stat_df[display_cols].to_string(index=False))

# — SARIMA order recommendation —————

print()
print('=' * 70)
print('SARIMA DIFFERENCING RECOMMENDATION')
print('=' * 70)

for period_label, _ in stationarity_periods:
    sub = stat_df[stat_df['Period'] == period_label]

    def verdict_for(trans, sub=sub):
        row = sub[sub['Transformation'] == trans]
        return row['Joint verdict'].values[0] if len(row) else 'N/A'

    raw_v = verdict_for('Raw')
    d1_v = verdict_for('Diff(1)')
    d12_v = verdict_for('Diff(12)')
    d1d12_v = verdict_for('Diff(1)+Diff(12)')

    # infer d
    if raw_v == 'Stationary':
        d = 0
    elif d1_v == 'Stationary':
        d = 1
    else:

```

```

    d = 2

    # infer D (seasonal differencing, period=12)
    D = 0 if raw_v == 'Stationary' else 1

    print(f'\n{period_label}')
    print(f'  Raw          → {raw_v}')
    print(f'  Diff(1)       → {d1_v}')
    print(f'  Diff(12)      → {d12_v}')
    print(f'  Diff(1)+Diff(12) → {d1d12_v}')
    print(f'  Suggested: d={d}, D={D}, s=12 → SARIMA(p, {d}, q)(P, {D}, Q)[12]')

# — visual: heatmap of ADF / KPSS p-values —————

plot_stationarity_heatmaps(
    stat_df,
    trans_order,
    period_label_order,
    'Stationarity p-values: ADF (left) and KPSS (right)',
)

```

STATIONARITY TEST RESULTS – ADF + KPSS

ADF H0: unit root (non-stationary) → reject ($p < 0.05$) = stationary

KPSS H0: stationary → reject ($p < 0.05$) = non-stationary

	Period	Transformation	ADF p	ADF stationary	KPSS p	KPSS stationary
stationary	Joint verdict					
	Full (2000–present)	Raw	0.1356	X	0.1000	
✓	Ambiguous					
	Full (2000–present)	Diff(1)	0.0000	✓	0.1000	
✓	Stationary					
	Full (2000–present)	Diff(12)	0.0051	✓	0.1000	
✓	Stationary					
	Full (2000–present)	Diff(1)+Diff(12)	0.0000	✓	0.1000	
✓	Stationary					
	Pre-Pandemic (2000–Mar 2020)	Raw	0.6776	X	0.0100	
X	Non-stationary					
	Pre-Pandemic (2000–Mar 2020)	Diff(1)	0.0000	✓	0.1000	
✓	Stationary					
	Pre-Pandemic (2000–Mar 2020)	Diff(12)	0.0004	✓	0.1000	
✓	Stationary					
	Pre-Pandemic (2000–Mar 2020)	Diff(1)+Diff(12)	0.0000	✓	0.1000	
✓	Stationary					
	Pandemic (Apr 2020–2021)	Raw	0.9991	X	0.0323	
X	Non-stationary					
	Pandemic (Apr 2020–2021)	Diff(1)	0.0498	✓	0.1000	
✓	Stationary					
	Pandemic (Apr 2020–2021)	Diff(12)	0.8644	X	0.0397	
X	Non-stationary					
	Pandemic (Apr 2020–2021)	Diff(1)+Diff(12)	0.0012	✓	0.0417	
X	Ambiguous					
	Post-Pandemic (2022+)	Raw	0.8104	X	0.0508	
✓	Ambiguous					
	Post-Pandemic (2022+)	Diff(1)	0.0000	✓	0.1000	
✓	Stationary					
	Post-Pandemic (2022+)	Diff(12)	0.0024	✓	0.1000	
✓	Stationary					
	Post-Pandemic (2022+)	Diff(1)+Diff(12)	0.0001	✓	0.1000	
✓	Stationary					

SARIMA DIFFERENCING RECOMMENDATION

Full (2000–present)

Raw → Ambiguous

Diff(1) → Stationary

Diff(12) → Stationary

Diff(1)+Diff(12) → Stationary

Suggested: $d=1, D=1, s=12 \rightarrow \text{SARIMA}(p, 1, q)(P, 1, Q)[12]$

Pre-Pandemic (2000–Mar 2020)

Raw → Non-stationary

Diff(1) → Stationary

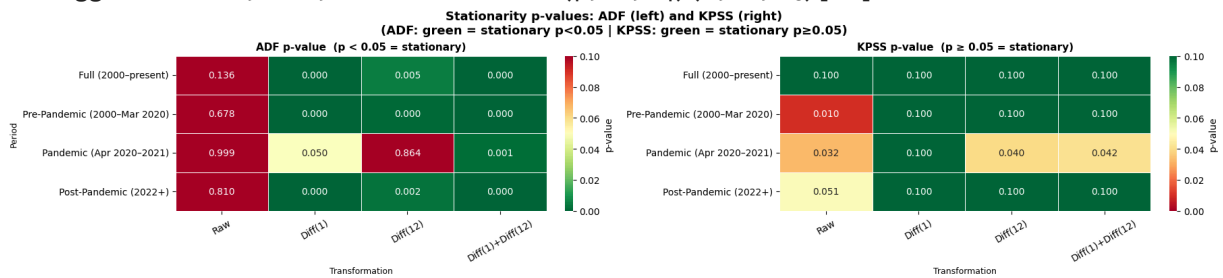
Diff(12) → Stationary
 Diff(1)+Diff(12) → Stationary
 Suggested: d=1, D=1, s=12 → SARIMA(p, 1, q)(P, 1, Q)[12]

Pandemic (Apr 2020–2021)

Raw → Non-stationary
 Diff(1) → Stationary
 Diff(12) → Non-stationary
 Diff(1)+Diff(12) → Ambiguous
 Suggested: d=1, D=1, s=12 → SARIMA(p, 1, q)(P, 1, Q)[12]

Post-Pandemic (2022+)

Raw → Ambiguous
 Diff(1) → Stationary
 Diff(12) → Stationary
 Diff(1)+Diff(12) → Stationary
 Suggested: d=1, D=1, s=12 → SARIMA(p, 1, q)(P, 1, Q)[12]



Auto correlation

```
In [73]: acf_periods = [
    'Pre-Pandemic (2000–Mar 2020)',
    'Pandemic (Apr 2020–2021)',
    'Post-Pandemic (2022+)',
]

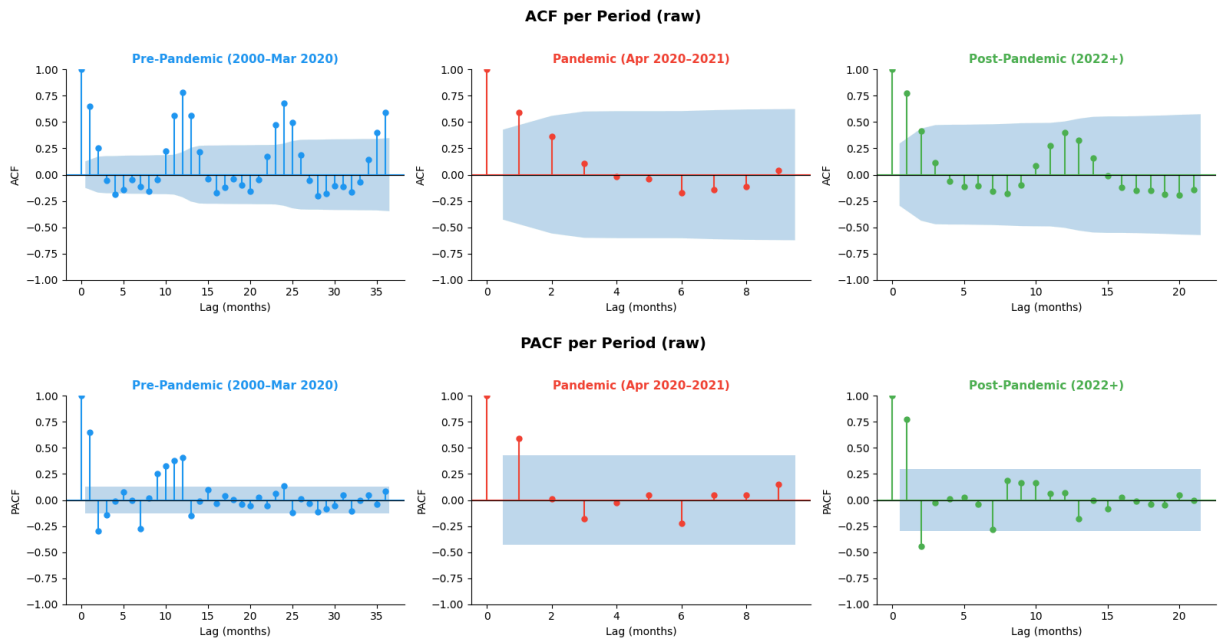
for plot_fn, fn_label in [(plot_acf, 'ACF'), (plot_pacf, 'PACF')]:
    fig, axes = plt.subplots(1, 3, figsize=(16, 4))
    fig.suptitle(
        f'{fn_label} per Period (raw)',
        fontsize=14,
        fontweight='bold',
        y=1.02,
    )

    for ax, period in zip(axes, acf_periods):
        series = monthly_df.loc[monthly_df['period'] == period, 'arrival_count']
        lags = min(36, len(series) // 2 - 1)
        color = period_colors[period]

        plot_fn(series, lags=lags, ax=ax, color=color, vlines_kwargs={'colors': col

        ax.set_title(period, fontsize=11, fontweight='bold', color=color)
        ax.set_xlabel('Lag (months)', fontsize=10)
        ax.set_ylabel(fn_label, fontsize=10)
        ax.axhline(0, color='black', linewidth=0.8)
        sns.despine(ax=ax)
```

```
plt.tight_layout()
plt.show()
```



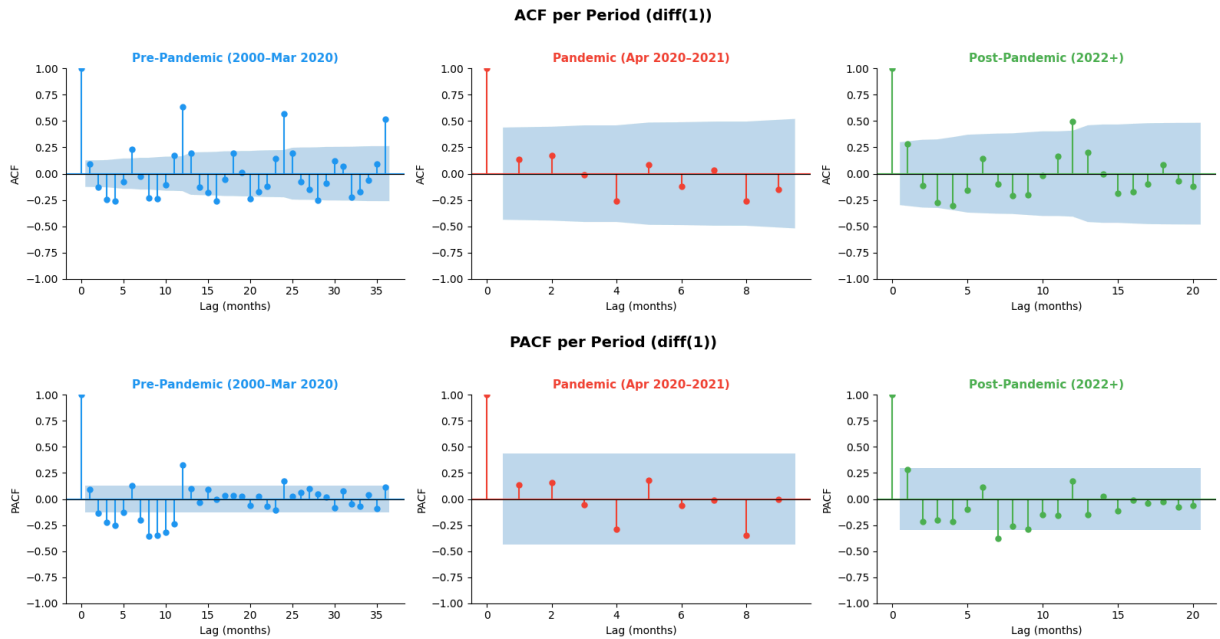
```
In [74]: for plot_fn, fn_label in [(plot_acf, 'ACF'), (plot_pacf, 'PACF')]:
fig, axes = plt.subplots(1, 3, figsize=(16, 4))
fig.suptitle(
    f'{fn_label} per Period (diff(1))',
    fontsize=14,
    fontweight='bold',
    y=1.02,
)

for ax, period in zip(axes, acf_periods):
    series = (
        monthly_df
        .loc[monthly_df['period'] == period, 'arrival_count']
        .diff(1)
        .dropna()
    )
    lags = min(36, len(series) // 2 - 1)
    color = period_colors[period]

    plot_fn(series, lags=lags, ax=ax, color=color, vlines_kwargs={'colors': col

    ax.set_title(period, fontsize=11, fontweight='bold', color=color)
    ax.set_xlabel('Lag (months)', fontsize=10)
    ax.set_ylabel(fn_label, fontsize=10)
    ax.axhline(0, color='black', linewidth=0.8)
    sns.despine(ax=ax)

plt.tight_layout()
plt.show()
```



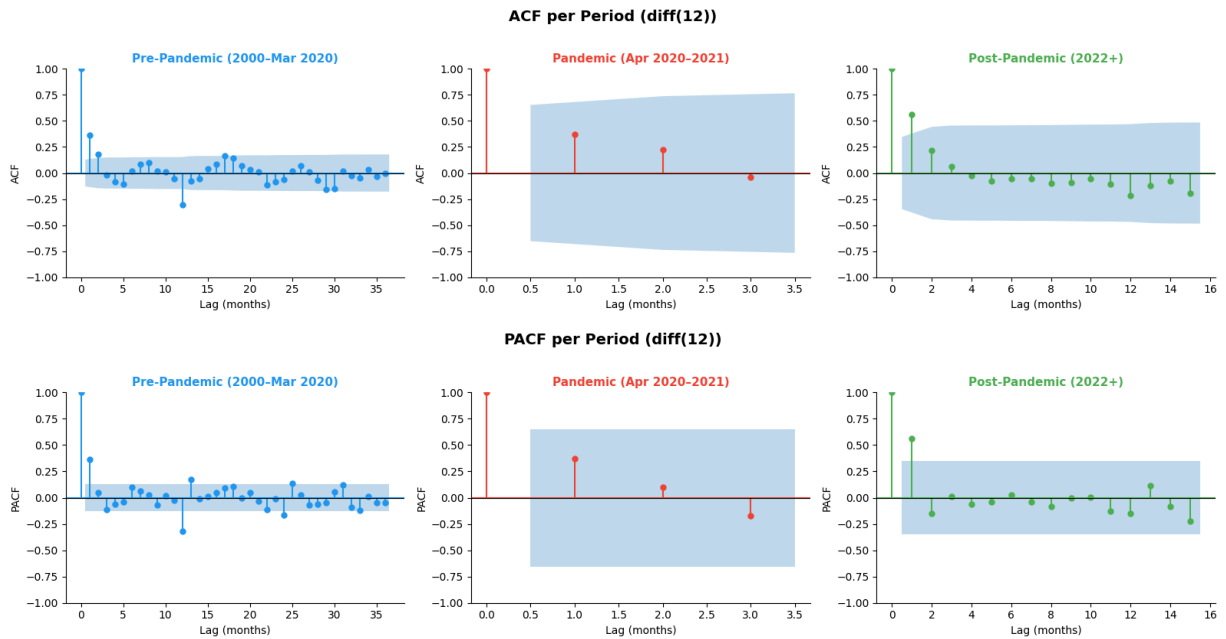
```
In [75]: for plot_fn, fn_label in [(plot_acf, 'ACF'), (plot_pacf, 'PACF')]:
fig, axes = plt.subplots(1, 3, figsize=(16, 4))
fig.suptitle(
    f'{fn_label} per Period (diff(12))',
    fontsize=14,
    fontweight='bold',
    y=1.02,
)

for ax, period in zip(axes, acf_periods):
    series = (
        monthly_df
        .loc[monthly_df['period'] == period, 'arrival_count']
        .diff(12)
        .dropna()
    )
    lags = min(36, len(series) // 2 - 1)
    color = period_colors[period]

    plot_fn(series, lags=lags, ax=ax, color=color, vlines_kwargs={'colors': col

    ax.set_title(period, fontsize=11, fontweight='bold', color=color)
    ax.set_xlabel('Lag (months)', fontsize=10)
    ax.set_ylabel(fn_label, fontsize=10)
    ax.axhline(0, color='black', linewidth=0.8)
    sns.despine(ax=ax)

plt.tight_layout()
plt.show()
```



Heteroscedasticity

A series is heteroscedastic when its variance changes over time. For tourism arrivals this usually shows up as a seasonal amplitude that grows with the level of the series. This is what the mean-std check in the *Trend* section suggests. It matters because ARIMA-type models assume constant error variance. Level-dependent variance also makes the prediction intervals too narrow at the peaks and too wide at the troughs.

Two complementary checks:

1. **Visual:** 12-month rolling mean, rolling standard deviation and rolling coefficient of variation (std / mean). A constant CV together with a growing std points to multiplicative behaviour.
2. **White test** via `check_heteroscedasticity` (Joseph & Tackes, *Modern Time Series Forecasting with Python 2E*, `src/transforms/stationary_utils.py`). The helper fits a linear trend and runs White's test on the residuals (H_0 : homoscedastic). It is applied to the raw series and to the series after linear detrending and period-average deseasonalizing (`DetrendingTransformer` → `DeseasonalizingTransformer`, same source). This way, the seasonal cycle is not mistaken for changing variance.

```
In [76]: fig, axes = plt.subplots(3, 1, figsize=(14, 10), sharex=True, constrained_layout=True)
fig.suptitle(
    '12-month Rolling Mean, Std and Coefficient of Variation (2000-present)',
    fontsize=14,
    fontweight='bold',
)

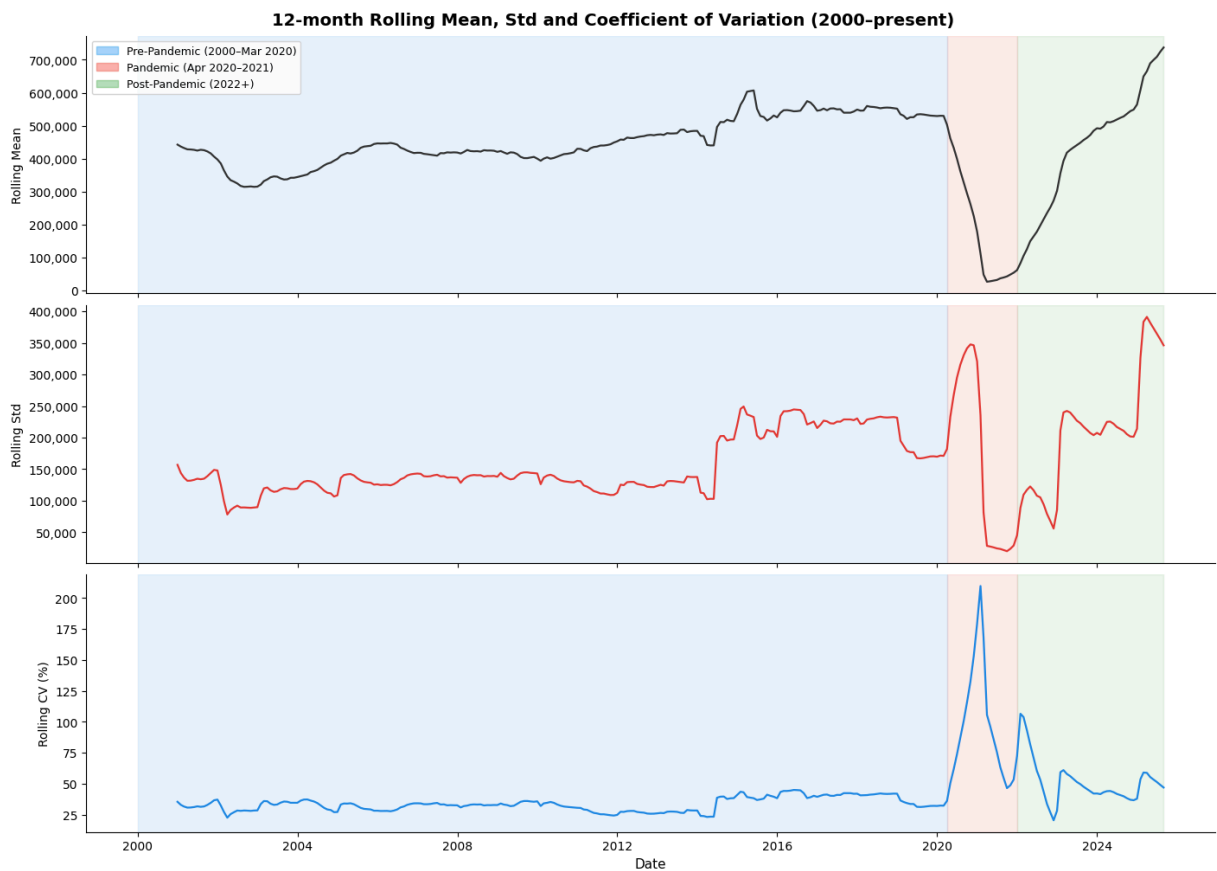
full_series = period_series['Full Period (2000-present)']
rolling_mean = full_series.rolling(12).mean()
rolling_std = full_series.rolling(12).std()
rolling_cv = rolling_std / rolling_mean * 100
```

```

for ax, values, label, color in [
    (axes[0], rolling_mean, 'Rolling Mean', '#333333'),
    (axes[1], rolling_std, 'Rolling Std', '#e53935'),
    (axes[2], rolling_cv, 'Rolling CV (%)', '#1e88e5'),
]:
    ax.plot(values.index, values, color=color, linewidth=1.6)
    for period, start, end in trend_periods:
        ax.axvspan(start, end, alpha=0.1, color=period_colors[period])
    ax.set_ylabel(label, fontsize=10)
    ax.yaxis.set_major_formatter(plt.FuncFormatter(lambda x, _: f'{x:,.0f}'))
    sns.despine(ax=ax)

axes[0].legend(
    handles=[
        mpatches.Patch(color=period_colors[p], alpha=0.4, label=p)
        for p, _, _ in trend_periods
    ],
    fontsize=9,
    frameon=True,
    loc='upper left',
)
axes[-1].set_xlabel('Date', fontsize=11)
plt.show()

```



```

In [77]: # Helpers `check_heteroscedasticity`, `DetrendingTransformer` and
# `DeseasonalizingTransformer` from Joseph & Tackes, "Modern Time Series Forecasting
# with Python" 2E - src/transforms/

```

```

def detrend_deseasonalize(series):
    """Linear detrend + period-average deseasonalize (AutoStationary order)."""
    detrended = DetrendingTransformer(degree=1).fit_transform(series, freq=FREQ)
    if len(detrended) <= 2 * SEASONAL_PERIOD:
        return detrended
    return DeseasonalizingTransformer(
        seasonal_period=SEASONAL_PERIOD, seasonality_extraction='period_averages'
    ).fit_transform(detrended, freq=FREQ)

def white_row(series, variant):
    res = check_heteroscedasticity(series, confidence=0.05)
    return {
        'Variant': variant,
        'White LM stat': round(res.lm_statistic, 3),
        'White LM p': round(res.lm_p_value, 4),
        'Heteroscedastic': '✓' if res.heteroscedastic else 'X',
    }

hetero_rows = []
for label, series in period_series.items():
    hetero_rows.append({'Period': label, **white_row(series, 'Raw')})
    hetero_rows.append({
        'Period': label,
        **white_row(detrend_deseasonalize(series), 'Detrended + deseasonalized'),
    })

hetero_tests = pd.DataFrame(hetero_rows).set_index(['Period', 'Variant'])

print('=' * 90)
print('WHITE TEST - H0: homoscedastic → reject (LM and F p < 0.05) = heteroscedastic')
print('=' * 90)
print(hetero_tests.to_string())

```

=====			
=====			
WHITE TEST – H0: homoscedastic → reject (LM and F p < 0.05) = heteroscedastic			
=====			
=====			
		White LM stat	White LM p H
eteroscedastic			
Period	Variant		
Full Period (2000–present)	Raw	35.552	0.0000
✓			
	Detrended + deseasonalized	39.978	0.0000
✓			
Pre-Pandemic (2000–Mar 2020)	Raw	10.891	0.0043
✓			
	Detrended + deseasonalized	3.386	0.1840
X			
Pandemic (Apr 2020–2021)	Raw	5.731	0.0570
X			
	Detrended + deseasonalized	5.731	0.0570
X			
Post-Pandemic (2022+)	Raw	3.452	0.1780
X			
	Detrended + deseasonalized	9.043	0.0109
✓			

Box-Cox Transformation (Guerrero λ)

Instead of fixing $\lambda = 0$ (log), the Box-Cox power transformation

$$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \log y, & \lambda = 0 \end{cases}$$

is fitted with λ chosen by **Guerrero's method** (Guerrero, 1993, *Journal of Forecasting* 12, 37–48). The series is split into consecutive seasonal blocks ($m = 12$, i.e. calendar years). Guerrero picks the λ that makes $\sigma_h / \mu_h^{1-\lambda}$ as constant as possible across blocks, measured by its coefficient of variation. In other words, λ is the power that best removes the dependence between the level and the spread.

The transformation uses `BoxCoxTransformer(seasonal_period=12, optimization='guerrero', bounds=(-1, 2))` from Joseph & Tackes, *Modern Time Series Forecasting with Python* 2E (`src/transforms/target_transformations.py`). The Guerrero optimizer is adapted by the authors from `sktime`.

Reading λ : ≈ 1 no transformation needed · ≈ 0.5 square root · ≈ 0 log · < 0 inverse power (stronger than log).

```
In [78]: # Helper `BoxCoxTransformer` (Guerrero λ) from Joseph & Tackes, "Modern Time Series
# Forecasting with Python" 2E – src/transforms/target_transformations.py

def interpret_lambda(lmbda):
```

```

if lmbda > 0.75:
    return '≈ none ( $\lambda \approx 1$ )'
if lmbda > 0.25:
    return '≈ square root ( $\lambda \approx 0.5$ )'
if lmbda > -0.25:
    return '≈ log ( $\lambda \approx 0$ )'
return 'inverse power ( $\lambda < 0$ )'

boxcox_fitted = {}
boxcox_series = {}
lambda_rows = []
for label, series in period_series.items():
    bc = BoxCoxTransformer(
        seasonal_period=SEASONAL_PERIOD, optimization='guerrero', bounds=(-1, 2)
    )
    bc.fit(series)
    boxcox_fitted[label] = bc
    boxcox_series[label] = bc.transform(series)
    lambda_rows.append({
        'Period': label,
        'n': len(series),
        'Guerrero blocks (years)': len(series) // SEASONAL_PERIOD,
        'λ (Guerrero)': round(bc.boxcox_lambda, 4),
        'Interpretation': interpret_lambda(bc.boxcox_lambda),
    })

boxcox_lambdas = pd.DataFrame(lambda_rows).set_index('Period')
print(boxcox_lambdas.to_string())

# — raw vs Box-Cox per period —————
fig, axes = plt.subplots(4, 2, figsize=(15, 14), constrained_layout=True)
fig.suptitle(
    'Raw Series vs Box-Cox (Guerrero λ) Transformed Series',
    fontsize=14,
    fontweight='bold',
)

for row, (label, series) in enumerate(period_series.items()):
    color = period_series_colors[label]
    lmbda = boxcox_fitted[label].boxcox_lambda

    axes[row, 0].plot(series.index, series, color=color, linewidth=1.2)
    axes[row, 0].set_title(
        f'{label} - Raw', fontsize=10, fontweight='bold', color=color
    )
    axes[row, 0].set_ylabel('Arrival Count', fontsize=9)
    axes[row, 0].yaxis.set_major_formatter(
        plt.FuncFormatter(lambda x, _: f'{int(x):,}')
    )

    transformed = boxcox_series[label]
    axes[row, 1].plot(transformed.index, transformed, color=color, linewidth=1.2)
    axes[row, 1].set_title(
        f'{label} - Box-Cox (λ = {lmbda:.3f})',
        fontsize=10,

```

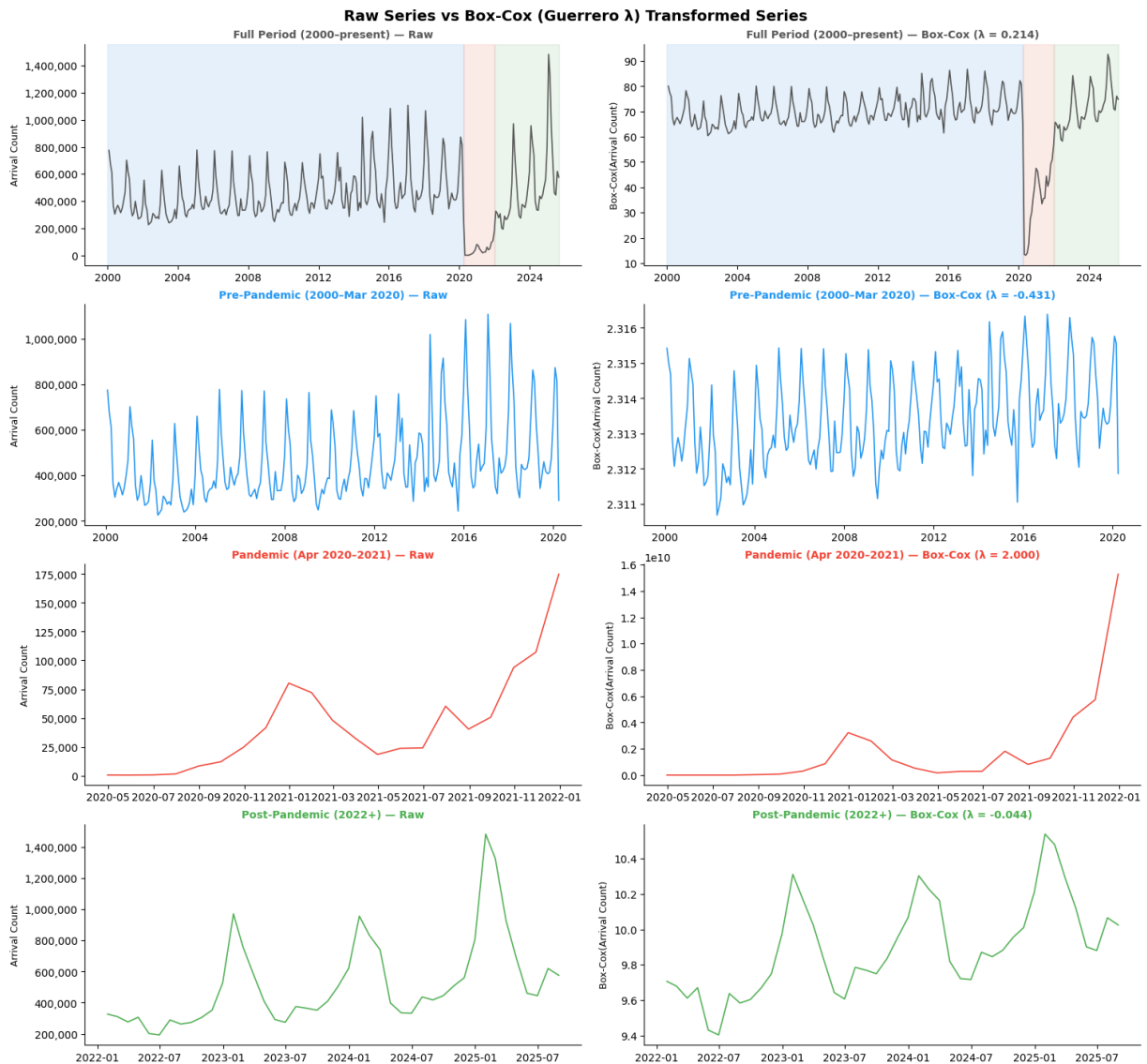
```
        fontweight='bold',
        color=color,
    )
    axes[row, 1].set_ylabel('Box-Cox(Arrival Count)', fontsize=9)

    if label == 'Full Period (2000-present)':
        for period, start, end in trend_periods:
            for ax in axes[row]:
                ax.axvspan(start, end, alpha=0.1, color=period_colors[period])

    for ax in axes[row]:
        sns.despine(ax=ax)

plt.show()
```

	n	Guerrero blocks (years)	λ (Guerrero)	Int
erpretation				
Period				
Full Period (2000-present)	308	25	0.2135	$\approx$
log ($\lambda \approx 0$)				
Pre-Pandemic (2000-Mar 2020)	243	20	-0.4306	inverse po
wer ($\lambda < 0$)				
Pandemic (Apr 2020-2021)	21	1	2.0000	$\approx n$
one ($\lambda \approx 1$)				
Post-Pandemic (2022+)	44	3	-0.0444	$\approx$
log ($\lambda \approx 0$)				



Did Box-Cox stabilize the variance?

The same diagnostics are repeated on the Box-Cox scale:

- the annual mean–std correlation from the additive/multiplicative check (computed on the level series), and
- the White test, run on both scales **after** detrending and deseasonalizing (`detrend_deseasonalize` above).

The Pandemic period has only two calendar years, so its mean–std correlation is trivially ± 1 and its Guerrero λ rests on two blocks. Treat that column as indicative only.

```
In [79]: def mean_std_corr(series):
          annual = series.resample('YE').agg(['mean', 'std']).dropna()
          return annual['mean'].corr(annual['std'])

          variance_rows = []
          for label, series in period_series.items():
```

```

transformed = boxcox_series[label]
raw_white = check_heteroscedasticity(detrend_deseasonalize(series))
bc_white = check_heteroscedasticity(detrend_deseasonalize(transformed))
variance_rows.append({
    'Period': label,
    'λ': round(boxcox_fitted[label].boxcox_lambda, 3),
    'Mean-std corr (raw)': round(mean_std_corr(series), 3),
    'Mean-std corr (Box-Cox)': round(mean_std_corr(transformed), 3),
    'White p (raw scale)': round(raw_white.lm_p_value, 4),
    'White p (Box-Cox scale)': round(bc_white.lm_p_value, 4),
    'Heteroscedastic (raw scale)': '✓' if raw_white.heteroscedastic else 'X',
    'Heteroscedastic (Box-Cox scale)': '✓' if bc_white.heteroscedastic else 'X'
})

variance_comparison = pd.DataFrame(variance_rows).set_index('Period')
print(variance_comparison.T.to_string())

```

Period	Full Period (2000–present)	Pre-Pandemic (2000–Mar 20)	Pandemic (Apr 2020–2021)	Post-Pandemic (2022+)
λ	0.214			-0.
431	2.0		-0.044	
Mean-std corr (raw)	0.597			0.
897	1.0		0.992	
Mean-std corr (Box-Cox)	-0.643			0.
358	1.0		0.966	
White p (raw scale)	0.0			0.
184	0.057		0.0109	
White p (Box-Cox scale)	0.0			0.2
633	0.0253		0.0014	
Heteroscedastic (raw scale)	✓			
X	X		✓	
Heteroscedastic (Box-Cox scale)	✓			
X	✓		✓	

Stationarity on the Box-Cox Scale

The ADF + KPSS grid from the *Stationarity* section is re-run on the Box-Cox transformed series. Each period uses its own Guerrero λ . This replaces the log-based variants that were previously in that grid.

```

In [80]: stationarity_label_map = dict(
    zip(period_label_order, period_series.keys())
) # 'Full (2000–present)' → 'Full Period (2000–present)'

def boxcox_for(series, period_label):
    return boxcox_fitted[stationarity_label_map[period_label]].transform(series)

boxcox_transformations = [
    ('Box-Cox', boxcox_for),
    ('Box-Cox + Diff(1)', lambda s, p: boxcox_for(s, p).diff(1)),
    ('Box-Cox + Diff(12)', lambda s, p: boxcox_for(s, p).diff(12)),
    ('Box-Cox + Diff(1)+Diff(12)', lambda s, p: boxcox_for(s, p).diff(1).diff(12)),
]

```

```

stat_df_boxcox = stationarity_grid(stationarity_periods, boxcox_transformations)

print('=' * 100)
print('STATIONARITY TEST RESULTS – Box-Cox (Guerrero  $\lambda$ ) variants')
print('=' * 100)
print(stat_df_boxcox[display_cols].to_string(index=False))

# side-by-side verdicts: raw vs Box-Cox
verdicts = pd.concat([stat_df, stat_df_boxcox]).pivot_table(
    index='Period', columns='Transformation', values='Joint verdict', aggfunc='first'
)
verdict_order = [
    'Raw',
    'Box-Cox',
    'Diff(1)',
    'Box-Cox + Diff(1)',
    'Diff(12)',
    'Box-Cox + Diff(12)',
    'Diff(1)+Diff(12)',
    'Box-Cox + Diff(1)+Diff(12)',
]
print()
print('JOINT VERDICT – raw vs Box-Cox')
print(verdicts.reindex(index=period_label_order, columns=verdict_order).T.to_string())

plot_stationarity_heatmaps(
    stat_df_boxcox,
    [t for t, _ in boxcox_transformations],
    period_label_order,
    'Stationarity p-values on the Box-Cox scale: ADF (left) and KPSS (right)',
)

```

STATIONARITY TEST RESULTS – Box-Cox (Guerrero λ) variants

	Period	Transformation	ADF p	ADF stationary	KPSS
p KPSS stationary	Joint verdict				
	Full (2000-present)	Box-Cox	0.0255	✓	0.10
00	✓ Stationary				
	Full (2000-present)	Box-Cox + Diff(1)	0.0000	✓	0.10
00	✓ Stationary				
	Full (2000-present)	Box-Cox + Diff(12)	0.0031	✓	0.10
00	✓ Stationary				
	Full (2000-present)	Box-Cox + Diff(1)+Diff(12)	0.0000	✓	0.10
00	✓ Stationary				
	Pre-Pandemic (2000-Mar 2020)	Box-Cox	0.5585	X	0.01
00	X Non-stationary				
	Pre-Pandemic (2000-Mar 2020)	Box-Cox + Diff(1)	0.0000	✓	0.10
00	✓ Stationary				
	Pre-Pandemic (2000-Mar 2020)	Box-Cox + Diff(12)	0.0022	✓	0.10
00	✓ Stationary				
	Pre-Pandemic (2000-Mar 2020)	Box-Cox + Diff(1)+Diff(12)	0.0000	✓	0.10
00	✓ Stationary				
	Pandemic (Apr 2020-2021)	Box-Cox	1.0000	X	0.03
46	X Non-stationary				
	Pandemic (Apr 2020-2021)	Box-Cox + Diff(1)	0.9935	X	0.06
36	✓ Ambiguous				
	Pandemic (Apr 2020-2021)	Box-Cox + Diff(12)	0.9989	X	0.05
18	✓ Ambiguous				
	Pandemic (Apr 2020-2021)	Box-Cox + Diff(1)+Diff(12)	0.9985	X	0.08
20	✓ Ambiguous				
	Post-Pandemic (2022+)	Box-Cox	0.4276	X	0.03
73	X Non-stationary				
	Post-Pandemic (2022+)	Box-Cox + Diff(1)	0.0002	✓	0.10
00	✓ Stationary				
	Post-Pandemic (2022+)	Box-Cox + Diff(12)	0.6812	X	0.10
00	✓ Ambiguous				
	Post-Pandemic (2022+)	Box-Cox + Diff(1)+Diff(12)	0.0000	✓	0.08
24	✓ Stationary				

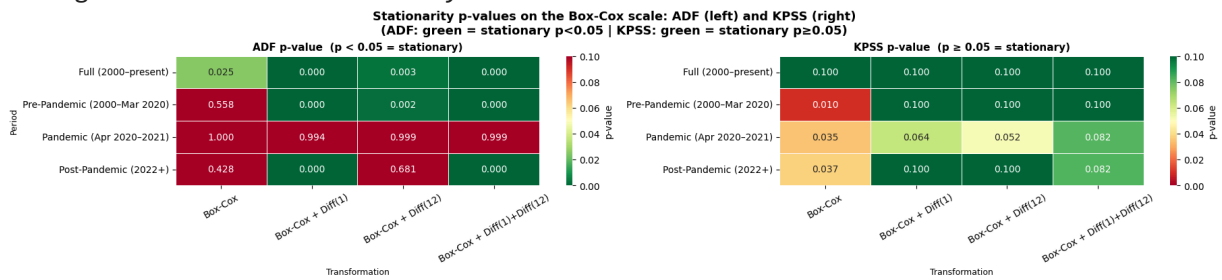
JOINT VERDICT – raw vs Box-Cox

Period Full (2000-present) Pre-Pandemic (2000-Mar 2020) Pandemic (Apr 2020-2021) Post-Pandemic (2022+)

Transformation

Raw		Ambiguous	Non-stationary
Non-stationary		Ambiguous	
Box-Cox		Stationary	Non-stationary
Non-stationary	Non-stationary		
Diff(1)		Stationary	Stationary
Stationary	Stationary		
Box-Cox + Diff(1)		Stationary	Stationary
Ambiguous	Stationary		
Diff(12)		Stationary	Stationary
Non-stationary	Stationary		
Box-Cox + Diff(12)		Stationary	Stationary
Ambiguous	Ambiguous		

Diff(1)+Diff(12)		Stationary		Stationary
Ambiguous		Stationary		
Box-Cox + Diff(1)+Diff(12)		Stationary		Stationary
Ambiguous		Stationary		



Automatic Stationarity Transformation (AutoML approach)

As a last step, `AutoStationaryTransformer` (Joseph & Tackes, *Modern Time Series Forecasting with Python 2E*, `src/transforms/target_transformations.py`) is used to let a rule-based pipeline choose the transformations. During `fit`, it runs the same tests used above, in this order:

1. **Trend:** `check_trend` (Mann-Kendall here). If a trend is found → `DetrendingTransformer(degree=1)`.
2. **Seasonality:** `check_seasonality` at $m = 12$. If seasonal and $n > 2m$ → `DeseasonalizingTransformer` (period averages).
3. **Heteroscedasticity:** `check_heteroscedasticity` (White). If heteroscedastic → `AddMTTransformer` when needed to make values positive, then `BoxCoxTransformer(optimization='guerrero')`.

The output is then checked with the ADF + KPSS pair from the *Stationarity* section. The fitted pipeline must also invert back to the original series (round-trip error), since forecasts will have to be mapped back to arrival counts.

Note that the pipeline removes a **deterministic** linear trend. It does not difference the series, so a stochastic trend (unit root) can remain in the output.

```
In [81]: # Helper `AutoStationaryTransformer` from Joseph & Tackes, "Modern Time Series
# Forecasting with Python" 2E - src/transforms/target_transformations.py

auto_transformers = {}
auto_outputs = {}
auto_rows = []
for label, series in period_series.items():
    # fresh dicts every fit: the transformer mutates its parameter dicts in place
    ast = AutoStationaryTransformer(
        confidence=0.05,
        seasonal_period=SEASONAL_PERIOD,
        trend_check_params={'mann_kendall': True},
        detrender_params={'degree': 1},
        deseasonalizer_params={},
        box_cox_params={'optimization': 'guerrero'},
    )
```

```

with warnings.catch_warnings():
    warnings.simplefilter('ignore')
    transformed = ast.fit_transform(series, freq=FREQ)
auto_transformers[label] = ast
auto_outputs[label] = transformed

steps = [type(tr).__name__.replace('Transformer', '') for tr in ast._pipeline]
box_cox = [tr for tr in ast._pipeline if isinstance(tr, BoxCoxTransformer)]
roundtrip_error = (ast.inverse_transform(transformed) - series).abs().max()
adf_r, kpss_r, verdict = test_series(transformed, label)

trend_flag = '✓' if ast._trend_check['trend'] else 'X'
hetero_flag = '✓' if ast._hetero_check['heteroscedastic'] else 'X'
auto_rows.append({
    'Period': label,
    'Pipeline': ' → '.join(steps) if steps else '(none)',
    'Trend (MK p)': f'{trend_flag} ({ast._trend_check["p_value"]:.4f})',
    'Seasonal (m=12)': '✓' if ast._seasonality_check['seasonal'] else 'X',
    'Heteroscedastic (White p)': (
        f'{hetero_flag} ({ast._hetero_check["lm_p_value"]:.4f})'
    ),
    'Box-Cox λ': f'{box_cox[0].boxcox_lambda:.3f}' if box_cox else '-',
    'ADF p': round(adf_r['p'], 4),
    'KPSS p': round(kpss_r['p'], 4),
    'Joint verdict': verdict,
    'Inverse round-trip max |error|': f'{roundtrip_error:.2e}',
})

auto_summary = pd.DataFrame(auto_rows).set_index('Period')

print('=' * 100)
print('AUTO STATIONARY TRANSFORMER – fitted pipelines and stationarity of the output')
print('=' * 100)
print(auto_summary.T.to_string())

```

===== AUTO STATIONARY TRANSFORMER – fitted pipelines and stationarity of the output =====

Period	Full Period (2000–present)	Pre-Pa
ndemic (2000–Mar 2020)	Pandemic (Apr 2020–2021)	Post-Pandemic (2022+)
Pipeline	Detrending → Deseasonalizing → AddM → BoxCox	Detren
ding → Deseasonalizing	Detrending Deseasonalizing → AddM → BoxCox	
Trend (MK p)		✓ (0.0084)
✓ (0.0000)	✓ (0.0058)	X (0.1070)
Seasonal (m=12)		✓
✓	✓	✓
Heteroscedastic (White p)		✓ (0.0000)
X (0.1840)	X (0.0570)	✓ (0.0400)
Box-Cox λ		0.989
–	–	-1.000
ADF p		0.1011
0.0121	0.0994	0.0
KPSS p		0.1
0.1	0.1	0.1
Joint verdict		Ambiguous
Stationary	Ambiguous	Stationary
Inverse round-trip max error		1.40e-09
0.00e+00	0.00e+00	1.57e-04

```
In [82]: fig, axes = plt.subplots(4, 2, figsize=(15, 14), constrained_layout=True)
fig.suptitle(
    'AutoStationaryTransformer: Original vs Transformed Series',
    fontsize=14,
    fontweight='bold',
)

for row, (label, series) in enumerate(period_series.items()):
    color = period_series_colors[label]
    transformed = auto_outputs[label]
    pipeline = auto_summary.loc[label, 'Pipeline']

    axes[row, 0].plot(series.index, series, color=color, linewidth=1.2)
    axes[row, 0].set_title(
        f'{label} – Original', fontsize=10, fontweight='bold', color=color
    )
    axes[row, 0].set_ylabel('Arrival Count', fontsize=9)
    axes[row, 0].yaxis.set_major_formatter(
        plt.FuncFormatter(lambda x, _: f'{int(x):,}')
    )

    axes[row, 1].plot(transformed.index, transformed, color=color, linewidth=1.2)
    axes[row, 1].axhline(
        transformed.mean(), color='black', linewidth=0.8, linestyle='--'
    )
    axes[row, 1].set_title(
        f'{label} – {pipeline}', fontsize=9, fontweight='bold', color=color
    )
    axes[row, 1].set_ylabel('Transformed', fontsize=9)
    axes[row, 1].annotate(
```

```

f'ADF+KPSS: {auto_summary.loc[label, "Joint verdict"]}',
xy=(0.03, 0.95),
xycoords='axes fraction',
va='top',
fontsize=8.5,
bbox=dict(boxstyle='round,pad=0.3', fc='white', alpha=0.8),
)

for ax in axes[row]:
    sns.despine(ax=ax)

plt.show()

```

